

Matching and Local Labor Market Size in Mexico: The Role of Informality

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Abstract

We examine the relationship between local labor market (LLM) size and the quality of worker-firm matching in Mexico using a matched employer-employee dataset of formal private-sector workers. Estimating two-way fixed effects models of log-wage, we find evidence of a positive assortative matching at the aggregate level. On average, doubling a local labor market's size increases matching quality by four to seven percentage points. However, labor informality significantly mediates this relationship. Higher informality is associated with lower assortative matching strength and a weaker connection between matching and LLM size. In markets with high informality, we observe that high-performing workers tend to fail to pair with high-performing firms. Furthermore, in these high-informality contexts, increased LLM size fails to translate into the expected matching advantages. Our results provide evidence suggesting that informality acts as a structural friction that prevents the realization of gains typically associated with urban scale in advanced economies.

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1 Introduction

Larger local labor markets (LLMs) have the potential to facilitate better matching between workers and employers. Intuitively, larger labor markets fuel a virtuous cycle: improved matching enhances productivity and raises salaries, which in turn attract more skilled candidates and firms that demand these skills, further increasing the chances of high-quality workers and firms finding each other.

Extensive empirical research supports the existence of these matching externalities in advanced economies ([Ahlfeldt and Pietrostefani, 2019](#); [Baum-Snow and Pavan, 2012](#); [Dauth et al., 2022](#); [D’Costa and Overman, 2014](#); [Duranton and Puga, 2004](#); [Gould, 2007](#); [Moretti and Yi, 2024](#)). This study contributes by analyzing the relationship between local labor market size and assortative matching in Mexico, a country whose structural features differ sharply from those of other countries in which this relationship has been studied. We find a positive association between LLM size and worker-workplace matching in Mexico. We also find, however, that labor informality plays an important role in mediating this association. Higher labor informality is associated with weaker assortative matching and a weaker connection between matching and LLM size. In particular, for the markets with the highest labor informality in our sample, a bigger market size does not translate into better worker-firm matching.

The weakening of the relationship between labor market size and matching in markets with high labor informality is consistent with evidence of adverse effects of informality, such as slower urban development and lower rates of skill improvement among workers compared with places with low informality ([Jedwab et al., 2022](#); [Duranton, 2015](#)). Besides informality, agglomeration economies arising from worker-workplace matching may operate differently in developing economies such as Mexico than in developed economies for other reasons. For example, while wage elasticities with respect to urban density in developing economies are, on average, comparable to those observed in developed ones, these gains are often insufficient to outweigh disutilities such as pollution, crime, and unreliable travel times ([Grover et al., 2023](#)).¹ Similarly, poor-quality transport infrastructure increases urban fragmentation. It reduces the scope of agglomeration economies — not only due to matching but also to sharing and learning ([Baum-Snow et al.,](#)

¹Specifically, [Grover et al. \(2023\)](#) show that, in a meta-analysis corrected for publication bias, the elasticities of wages, labor productivity, and TFP with respect to density are not significantly higher in developing countries than in developed ones.

2017; Ghani et al., 2016). In general, the literature has found that the determinants of wage dispersion across places in developing economies differ from those in developed economies (Bassier, 2023; Diallo et al., 2022; Frías et al., 2022; Pérez Pérez and Nuño-Ledesma, 2024).

To evaluate the relationship between LLM size and worker-workplace sorting, we first construct a metric of assortative matching. To this end, we leverage data from Social Security records that comprise the near-universe of formal workers in Mexico from 2004 to 2018. With these records, we estimate models of log wages with additive worker and workplace fixed effects using the approach commonly referred to as AKM, first popularized by Abowd et al. (1999). To proxy assortative matching, we use the correlation between worker and workplace fixed effects at the local labor market level, following Card et al. (2013) and Dauth et al. (2022).

We proceed by taking our measure of assortative matching and correlating it with the population at the LLM level, defining LLMs as commuting zones or commuting-zone-industry cells. We find that the association between LLM size and the degree of positive assortative matching in formal labor markets in Mexico is similar to equivalent metrics from research about labor markets in Germany. At the commuting-zone level and after correcting for limited mobility bias in the AKM model, we estimate that the average assortative matching measure in Mexico is 0.196, compared to an (uncorrected for limited mobility bias) average of 0.164 in Germany (Dauth et al., 2022). Also, at this level, the slope of the relationship between log LLM size and our matching measure ranges from 0.025 to 0.068, close to the slope of 0.052 estimated by Dauth et al. (2022).²

We then examine the relationship between informality, education, and the extent of assortative matching, and between these characteristics and the relationship between matching and LLM size. We find that labor informality is a significant correlate of the degree of assortative matching. In markets with high labor informality, both the level of assortative matching and its relationship to LLM size are weaker. Markets with more educated populations also exhibit better matching, but this relationship disappears once we control for informality.

We contribute to three strands of literature. First, we contribute to the studies documenting the presence of city-size wage gaps not only in developed countries, as discussed by Baum-Snow and Pavan (2012); Gould (2007); D’Costa and Overman (2014), and De la Roca and Puga (2016), but

²This value corresponds to the slope estimated by Dauth et al. (2022) corrected for limited-mobility bias using 10 clusters (see Table B.3, panel B in their online appendix).

also in developing economies (Chauvin et al., 2017; Combes et al., 2020; De la Roca et al., 2023; Duranton, 2016). Second, we contribute to the literature on matching in labor markets (Andersson et al., 2007; Baum-Snow and Pavan, 2012; Behrens et al., 2014; Dauth et al., 2022) by showing that matching externalities are similar in Mexico and in developed economies and by finding that labor informality reduces positive assortative matching in formal labor markets. In doing so, we also contribute to the literature on informal labor markets (Ulyssea, 2018) and their relationship with formal ones (Levy Algazi, 2018; Ulyssea, 2010). Lastly, we contribute to the small but growing set of studies, such as Frías et al. (2022) and Pérez Pérez and Nuño-Ledesma (2024), that adapt multi-dimensional fixed-effects models to the Mexican case.

The rest of the paper proceeds as follows. Section 2 provides an overview of the data and the Mexican context. Section 3 details our AKM model estimation. Section 4 estimates the relationship between assortative matching and local labor market size in Mexico. Section 5 discusses potential mechanisms that may change that relationship, such as informality and education. Section 6 concludes.

2 Data

We use social security records from the Mexican Social Security Institute (IMSS, by its Spanish acronym). IMSS is the Mexican government’s paramount social security, pension, and public health administrator. Salaried workers in the private sector are legally required to register with IMSS. According to the government’s estimate, 83% of all formal workers are affiliated with IMSS (Pérez Pérez and Nuño-Ledesma, 2024). The records at our disposal report monthly observations from November 2004 to December 2018. The data for the last month covered approximately 20.1 million workers.³ We provide additional details about the IMSS data in Appendix A.

Our data are not without limitations. Public sector employees are not included in our data because a different agency manages their records. Wages for high-earning workers are censored.⁴

³ We rely on the *registro patronal* as our employer identifier because it is more precise than alternatives like the *Registro Federal de Contribuyentes*. Assigned by the social security administration, the former more accurately captures employment structures in settings where multiple employers operate within a single plant. *Registros* are anonymized by our data provider, and their masking procedures are inconsequential to our estimates. See online Appendix A for details on variable construction and descriptive statistics.

⁴ The censoring limit was 25 daily minimum wages before 2017 and 25 “units of measurement and update” after 2017. These thresholds amount to about 1,539 pesos per day, or approximately 80 USD at 2018 exchange rates.

Information on persons working in the informal economy is absent from IMSS data.⁵ When estimating the AKM models, the dependent variable of interest is the natural logarithm of real daily taxable income.⁶ Taxable income may include remunerations made to the worker other than wages.⁷ The dataset also includes information on gender, age, and registration date with the IMSS. The dataset does not include information on education or hours worked. Our primary sample of interest consists of prime-age men (25-54 years old) who have likely completed their education. Therefore, their estimated worker effects capture schooling, and the wage variance attributable to education is reflected in the variance of worker effects. Regarding employer information, the data includes workplace and economic sector identifiers.

We complement IMSS data with local labor market (LLM) characteristics. We calculate population using municipality-level data from Mexican censuses and intercensal surveys, and then we aggregate it at the LLM level.⁸ For this aggregation, we rely on the definition from [Aldeco et al. \(2024\)](#), which divides Mexico into 777 LLMs.⁹ We also obtain informality rates and mean years of schooling from [Aldeco et al. \(2024\)](#).

3 Constructing a Measure of Assortative Matching

Here we describe the methodology we used to construct our measure of assortative matching. Specifically, we measure assortative matching as the correlation between worker and workplace

The presence of top-coding may bias our results. Wage censoring will reduce the correlation between worker and firm fixed effects for firms with high fixed effects. If censoring is more common in large cities or among high-skill workers, we may underestimate matching in large cities. Only about 2% of workers have censored wages in our sample, so we might expect this bias to be small. Nevertheless, for robustness, we imputed wages for the censored observations following [Card et al. \(2013\)](#) and calculated the correlations between worker and firm fixed effects from AKM estimates using data that included imputed wages. The resulting firm-worker fixed-effect correlations are nearly unchanged, suggesting that our main results are robust to wage censoring. Appendix B provides details about this procedure.

⁵Informality is widespread in Mexico; according to the country’s National Survey of Occupation and Employment (ENOE by its acronym in Spanish), 55% of all workers operate in the informal economy ([Banco de México, 2023](#)).

⁶As [Dauth et al. \(2022\)](#) note, using nominal or real wages only changes the scale of the firms’ fixed effects.

⁷One may be concerned that firms manipulate compensation schemes and underreport wages. [Kumler et al. \(2020\)](#) show that wage underreporting has declined since 1997, due to the Mexican pension reform that tied pension benefits to reported wages. Additionally, [Puggioni et al. \(2022\)](#) argue that underreported wages should not have been a problem during the period they studied.

⁸We use the information from the closest census or intercensal survey for each time interval. For 2004-2008, we use the 2005 intercensal survey. For 2009-2013, we use the 2010 census. For 2014-2018, we use the 2020 census.

⁹[Aldeco et al. \(2024\)](#) calculate commuting zones using the methodology in [Fowler and Jensen \(2020\)](#). Appendix C provides details on the construction of the commuting zones.

fixed effects, estimated using log-wage models following the AKM framework [Abowd et al. \(1999\)](#), as in [Card et al. \(2013, 2018\)](#). In AKM models, log wages are modeled as a function of additive worker and workplace fixed effects:

$$\ln(wage_{it}) = \alpha_i + \psi_{J(i,t)} + X'_{it}\beta + r_{it}. \quad (1)$$

Here, $wage_{it}$ is the real wage of worker i at time t . The vector of fixed effects α_i captures the influence of all time-invariant worker characteristics. Similarly, the vector of fixed effects $\psi_{J(i,t)}$ collects time-invariant factors at the workplace level for workplace J where worker i was employed at time t .¹⁰ The vector X_{it} includes control variables, which in our estimations include functions of age and time-interval trends. We estimate equation (1) by OLS with a preconditioned iterative gradient method ([Card et al., 2013](#)).¹¹ We generate model estimates for three discrete time segments: 2004-2008, 2009-2013, and 2014-2018.¹² Generating results for three time segments allows us to better describe potential changes in the total-variance contributions made over time by each of the components because the set of attributes that remain time invariant in any given panel can change with the length of the panel, as pointed out by [Millimet and Bellemare \(2025\)](#).

In AKM models, worker mobility identifies firm and worker effects. As pointed out by [Abowd et al. \(1999\)](#), these effects can be disentangled by worker mobility –generated when workers change employers– which creates a network of directly or indirectly connected workplaces. We restrict estimation to the largest “connected set” of workplaces in each time interval across which workers change jobs at least once ([Abowd et al., 1999](#)).¹³

The AKM specification allows for a decomposition of wage variance as noted by [Card et al.](#)

¹⁰IMSS identifies workplaces using the *registro patronal* (employer registry number), which tracks employers rather than physical locations. Therefore, we refer to the “*registro patronal* component” of wage variance as the workplace component instead of a plant effect.

¹¹We provide validation exercises for the linearity in equation (1) and the uncorrelatedness of the error term r_{it} and the fixed effects and covariates in online Appendix D. Specifically, we follow [Card et al. \(2013\)](#) and show that firms are exchangeable: for a worker, moving from firm A to firm B has approximately the same effect on wages as moving from firm B to firm A. [Pérez Pérez and Nuño-Ledesma \(2024\)](#) also provide evidence of the validity of AKM models in this sample.

¹²Our periods cover fewer years than those in [Dauth et al. \(2022\)](#). However, identification issues are not a primary concern since we use monthly frequency data. Additionally, we conduct our regression analysis of city size and matching by pooling the periods and including period fixed effects.

¹³Online appendix table A.1 shows that the descriptive statistics are similar across the full and connected-set samples.

(2013):

$$\begin{aligned} \text{Var}(\ln \text{wage}_{it}) = & \underbrace{\text{Var}(\alpha_i)}_{\text{workers}} + \underbrace{\text{Var}(\psi_{\mathbf{J}(i,t)})}_{\text{workplaces}} + \text{Var}(x'_{it}\beta) + \text{Var}(r_{it}) \\ & + 2 \underbrace{\text{Cov}(\alpha_i, \psi_{\mathbf{J}(i,t)})}_{\text{Assortative Matching}} + 2\text{Cov}(\psi_{\mathbf{J}(i,t)}, x'_{it}\beta) + 2\text{Cov}(\alpha_i, x'_{it}\beta). \end{aligned} \quad (2)$$

This decomposition guides our measurement of assortative matching at the local labor market level. Following [Dauth et al. \(2022\)](#), the matching measure is the correlation between worker and firm fixed effects in each local labor market.

For this variance decomposition, a large literature has documented that bias in the estimated firm and worker effects can lead to biased estimates of the variance components. This bias arises when workers' mobility across firms is limited ([Bonhomme et al., 2019](#); [Jochmans and Weidner, 2019](#); [Kline et al., 2020](#)). When too few workers change firms, it is difficult to reliably identify workplace effects; as a result, idiosyncratic wage shocks are erroneously attributed to firm fixed effects, biasing the estimated variance of those effects upward. Although [Pérez Pérez and Nuño-Ledesma \(2024\)](#) show that correcting for this bias leads to small changes in variance components in this data at the national level, presumably due to the monthly frequency, measuring assortative matching at the local labor market level, where mobility is limited within particular labor markets, may bias our assortative matching measure. To address these potential biases, we apply the [Bonhomme et al. \(2019\)](#) correction, cluster firms into five groups using 20 percentiles of the within-firm wage distribution as clustering variables, and estimate the model with this reduced number of workplace-effect parameters.¹⁴

Table 1 provides summary statistics for the AKM model outcomes. The models account for approximately 93% of the variance in log wages across all periods. The table also shows an increasing role for matching in explaining wage variation in Mexico, with its variance share rising

¹⁴We use the [Bonhomme et al. \(2019\)](#) correction for two reasons. First, we diagnosed the bias in our estimation of the variance of workplace fixed effects using the method from [Jochmans and Weidner \(2019\)](#). We calculated the sum of the trace of the normalized Laplacian matrix for the connected set. The estimated bias in percentage terms is around 25.3%, 23.5%, and 23.8% for 2004-2008, 2009-2013, and 2014-2018, respectively. When using the [Bonhomme et al. \(2019\)](#) methodology, our estimates are 24.3%, 21.8%, and 20.9% smaller than the uncorrected estimates of the variance of workplace effects. These differences are close to the estimated bias obtained with the [Jochmans and Weidner \(2019\)](#) method. Second, the [Bonhomme et al. \(2019\)](#) correction is computationally more feasible than the correction proposed by [Kline et al. \(2020\)](#).

from 0.36 during 2004–2008 to 0.40 during 2014–2018. Nevertheless, this is still far below the 0.64 observed in Germany during the 2008–2014 period (Dauth et al., 2022).

Table 1: AKM Model Estimation Results

	Interval 1 2004-2008	Interval 2 2009-2013	Interval 3 2014-2018
Worker and workplace parameters			
Number of worker effects	11,363,073	13,083,589	15,512,438
Number of workplace effects	858,480	892,929	1,009,320
Summary of parameter estimates			
St. dev. of worker effects	0.534	0.515	0.502
St. dev. of workplace effects	0.400	0.435	0.446
Correlation worker/workplace effects	0.363	0.370	0.398
Correlation worker effects/Xb	-0.065	-0.026	-0.046
Correlation workplace effects/Xb	-0.004	0.004	0.001
Goodness of fit			
St. dev. of log wages	0.808	0.823	0.829
RMSE	0.214	0.218	0.221
R Squared	0.930	0.930	0.929
Adj. R Squared	0.928	0.927	0.927
Total variance shares			
Worker effects	0.436	0.392	0.367
Workplace effects	0.245	0.279	0.290
2Cov(worker effects, workplace effects)	0.237	0.245	0.260
Remainder	0.082	0.085	0.083

Source: Authors' calculations using IMSS data. Results from estimating equation (1) via OLS with a preconditioned gradient method following Card et al. (2013) and using the Bonhomme et al. (2019) correction, where workplaces are clustered into five groups based on within-workplace wage distributions before estimating the AKM model. The total variance shares are given by equation (2). The estimations are restricted to prime-aged men (ages 25-54) in the largest connected set of firm groups for each time interval. All the estimations include the following controls: age, age squared, age cubed, and a monthly time trend. RMSE is the root mean squared error.

The last rows of Table 1 show the results of this decomposition. Worker effects explained about 44% of wage variance in 2004-2008, and their contribution to total variance has decreased over time. In contrast, the contribution of workplace effects and sorting has been increasing over time. Sorting explains 24% of wage variance in 2004-2008, whereas it explains approximately 26% in 2014-2018. This pattern suggests that assortative matching is increasingly important in determining wage variance in Mexico over time.

4 Assortative Matching and Labor Market Size

We proceed to analyze how the strength of positive assortative matching varies with local labor market size across Mexico’s LLMs. To this end, we regress our matching measure against the logarithm of the population at the LLM level, following [Dauth et al. \(2022\)](#).¹⁵ We use two definitions of local labor markets: the commuting zones defined by [Aldeco et al. \(2024\)](#) and the intersection of commuting zones and industries. The second definition is appropriate if matching is sector-specific, such that highly productive workers tend to work at highly productive firms but tend to search for jobs within specific sectors.¹⁶ To contextualize our findings, we compare our estimates to previous estimations of equivalent metrics obtained for Germany by [Dauth et al. \(2022\)](#). Each panel in Figure 1 shows a scatterplot depicting the association between log-population and our matching metric at two levels of aggregation: commuting zone and commuting-zone-industry. The relationship aligns with the German estimates.

Further econometric analysis confirms that the link between market size and assortative matching is similar in Mexico to that in Germany at the commuting zone level. We pool our estimated matching measure in a single AKM model and include binary variables indicating the period accounted for in the regression.¹⁷ We show the results of our estimations in Table 2. Panel A, column 1, shows that the correlation between labor market size and assortative matching for Mexico’s commuting zones is positive and statistically significant. The link between matching and labor market size is similar to that found in [Dauth et al. \(2022\)](#) for Germany. Doubling the commuting-zone population increases our matching metric by 4.1 p.p. Column 2 shows estimates relying on commuting-zone-industry cells as the definition of LLM, yielding similar results.¹⁸

In the remaining panels (B-D) of Table 2, we present estimates from alternative specifications to gauge the robustness of our findings. To residualize firm fixed effects for Panel B, we regress them on industrial sector effects (NAICS 2-digit level). We then calculate the correlation between

¹⁵In Appendix Table E.1, we show that we obtain similar results using log-employment as a proxy for labor market size.

¹⁶This definition is similar to the one used by [Dauth et al. \(2022\)](#), who look at matching within city-occupation cells. We use a two-digit NAICS industry classification. Our data includes 22 industries. We exclude cells with fewer than 50 workers or fewer than five firms.

¹⁷Appendix table E.2 shows that we obtain similar results when we estimate these regressions separately for each time interval.

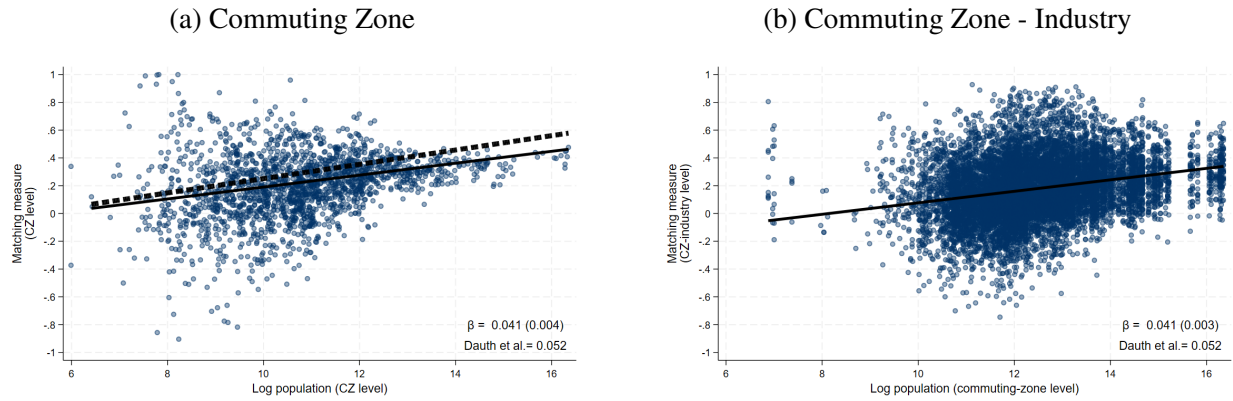
¹⁸For the commuting-zone-by-industry estimates, we restrict to cells with more than five firms and more than 50 workers, following [Dauth et al. \(2022\)](#).

Table 2: City Size and Assortative Matching in Mexico's Formal Labor Markets

Dependent variable: Matching measure - Correlation of worker and workplace FE		
	(1) CZ	(2) CZ-Industry
A: Baseline Model		
Log Population	0.0408*** (0.004)	0.0410*** (0.003)
R ²	0.083	0.070
B: Correlation of worker and residual workplace FE		
Log Population	0.0346*** (0.004)	0.0409*** (0.003)
R ²	0.065	0.070
C: Log population instrumented with population in 1921-1950		
Log Population	0.0309*** (0.008)	0.0432*** (0.010)
R ²	0.078	0.070
First-stage F	59.637	4.783
N for Panels A-C	1,913	10,097
Mean of dep. variable	0.196	0.180
D: Uncorrected for limited mobility bias		
Log Population	0.0679*** (0.004)	0.0531*** (0.004)
R ²	0.155	0.090
N	1,961	10,118
Mean of dep. variable	-0.014	0.000
E: Dropping the 10% largest and smallest areas		
Log Population	0.0250*** (0.007)	0.0438*** (0.004)
R ²	0.089	0.053
N	588	7,888
Mean of dep. variable	0.290	0.199

Source: Author's calculations using IMSS and INEGI data. "CZ" stands for commuting zone. The regressions pool data from 2004-2008, 2009-2013, and 2014-2018 with interval dummies. Column (2) restricts to cells with over five firms and 50 workers. Panels show the following estimates: A - baseline (Bonhomme et al. (2019)'s limited-mobility bias correction with five workplace clusters); B - industry-demeaned workplace fixed effects; C - we instrument population with historical population at the CZ level, relying on historical population estimates from Alix-Garcia and Sellars (2020). Log population instrumented with log population in 1921, 1930, 1940, and 1950 (Table I.1 shows different specifications of the historical population IV); D - no limited-mobility bias correction; and E - excluding extreme populations. Clustered standard errors at the commuting-zone level in parentheses. *: p<0.1, **: p<0.05, ***: p<0.01.

Figure 1: City Size and Assortative Matching in Mexico's Formal Labor Markets



Source: Author's calculations using IMSS and INEGI data. Each panel displays a scatterplot illustrating the relationship between log population and the assortative matching measure estimated from AKM models at the commuting zone and commuting zone industry levels. All estimates use the [Bonhomme et al. \(2019\)](#) correction, in which workplaces are clustered into five groups based on within-workplace wage distributions before estimating the AKM model. For comparison, Panel (a) displays the fitted line with the slope estimated for Germany by [Dauth et al. \(2022\)](#) corrected for limited-mobility bias (Table B3, Appendix B in [Dauth et al. \(2022\)](#)). For Panel (b), we restrict to cells with more than five firms and more than 50 workers. We classify industries using 2-digit NAICS codes. The bottom-right values show the slope of the linear regression for the displayed relationship. The regressions include dummies for each time interval. Clustered standard errors at the commuting-zone level in parentheses.

the residuals of the previous regression and the person fixed effects at each geographical level. This approach estimates correlations between city size and matching that control for different industry compositions across cities. Using these “residual” fixed effects, the association between city size and matching is slightly weaker than in previous panels. This pattern indicates that part of the effect seen in the estimates with unadjusted fixed effects was due to industries with better labor market matching being located in larger cities.

To avoid reverse causality between matching and demographic growth, we instrument the current population with populations in 1921, 1930, 1940, and 1950 as reported by [Alix-Garcia and Sellars \(2020\)](#). These historical estimates are based on 15-by-15-kilometer grid cells, which we aggregate to the desired levels. Panel C shows that this method yields a weaker matching-market size link than in Panel A, although the relationship remains significant.¹⁹

In Panel D, we report results without adjusting for limited mobility bias. These estimates are more directly comparable to the baseline results from [Dauth et al. \(2022\)](#). Without the correction, the association between LLM size and matching is stronger than in the baseline results corrected for limited mobility. We interpret this as evidence that mobility bias does not fundamentally alter the pattern we find.

Last, Panel E removes the largest and smallest areas from the sample. At the commuting zone level, we observe a substantial decrease in the estimate relative to the baseline. However, the results at the commuting zone-industry level are similar in this restricted sample to those in the full sample, implying that the small association we find is not due to the smallest commuting zones or to Mexico City’s influence.²⁰ Overall, the city-size advantage for matching in the labor market in Mexico is similar to the results of [Dauth et al. \(2022\)](#).

¹⁹ We provide details about historical population and data in Appendix I, and we show estimates instrumenting current population with historical population values from different years between 1921 and 1950 in Appendix Table I.1. More recent historical population values are stronger predictors of the current population. We have historical population data for 1900 and 1910. However, we excluded these years from our estimations because the Hansen test for over-identifying restrictions, when including these periods, rejected the validity of the instruments.

²⁰ Since the correlation coefficients between worker and firm fixed effects are themselves estimated, there may be heteroskedasticity in the regression model because in smaller local labor markets, the estimated correlation coefficient will have a larger sampling error. To check if this heteroskedasticity affects our results, we re-estimate the regressions in Table 2 using a non-parametric correction for heteroskedasticity ([Cadena, 2014](#)). The results are in Appendix Table E.3 and are similar to our baseline results.

5 Factors Limiting Matching Externalities from Larger Labor Markets in Mexico

We now assess potential factors that may mediate matching externalities in Mexico. At the local labor market level, we focus on the role of labor informality and differences in educational attainment. At the firm level, we focus on firm size and on differences in matching across industries.²¹

We choose these mechanisms since we deem them the most distinctive of Mexican labor markets relative to those in developed countries. Nevertheless, other mechanisms may also influence the extent of matching and its relationship to city size. For example, mobility frictions and spatial mismatch may prevent workers from reaching firms that are a good match. Information asymmetries may also prevent good matches when firms are unable to judge the quality of their workers.

5.1 Labor Informality

Labor market informality is high in Mexico. According to Mexico’s labor survey (ENOE), 54% of employed workers were informal as of April 2025. Informal workers may not pay income taxes and lack labor stability and access to social security through their employers. The high rate of labor informality has been documented as a limitation to productivity growth in Mexico ([Levy Algazi, 2018](#)).

We hypothesize that a large informal labor market may depress assortative matching in the formal labor market for multiple reasons, drawing on the study of agglomeration effects. The first reason may be increased search costs for firms seeking workers who are a good match. The reduction in search costs as economies grow larger is one of the traditional channels for agglomeration effects associated with city size ([Henderson, 1986](#)). However, informality may disrupt this relationship by attracting workers to informal jobs, decreasing formal matches, and increasing search costs for formal labor market firms ([Helsley and Strange, 1990](#)). A larger share of informal firms may make it harder to fill vacancies in the formal sector if it captures a large share of the labor supply ([Petrongolo and Pissarides, 2006](#)), or if it increases reservation wages for workers who use

²¹We also analyze the role of unions in Appendix Section F. Our analysis for Mexico suggests that while unions may substitute for market-based matching in some areas, strong unions generally flatten the relationship between city size and sorting by enforcing non-market job allocation rules.

informal wages to set reference wages against which to compare formal job offers.

Consistent with a scenario in which the benefits of acquiring specialized human capital are lower, higher-informality LLMs exhibit lower assortative matching and reduced city-size advantages relative to lower-informality LLMs. [Rotemberg and Saloner \(2000\)](#) argue that for workers to invest in specific human capital, they need to be able to recoup the investment cost afterward. While a larger availability of informal jobs may allow workers to experiment before specializing ([Wheeler, 2008](#); [Bleakley and Lin, 2012](#)), it may also deter them from specializing. Acquiring specific human capital may not be profitable for workers in markets where abundant informal labor opportunities do not require it. Consistent with this mechanism, in [Appendix L](#) we show a strong negative relationship between local informality rates and the share of high-skill workers (those with 16 or more years of education relative to lower-educated workers).

High labor informality may further diminish formal firms' motivation to seek optimal worker matches. If the prevalence of informal firms reduces the overall productivity of the formal sector, firms may be less inclined to invest in the search costs necessary for high-quality matches, even in large markets. This can result in formal firms hiring fewer high-fixed-effect workers or opting for informal hiring practices, thereby weakening the observable matching quality within the formal sector.

[Figure 2](#) shows the relationship between city size and assortative matching, separating the sample by quantiles of informality, using commuting-zone-level informality data from the economic censuses compiled by [Aldeco et al. \(2024\)](#). The Figure illustrates that higher informality is associated with lower assortative matching in the formal sector and a lack of matching advantages in large markets. For markets in the first three quartiles of informality, with informality rates below 72%, the relationship is similar to that in [Figure 1](#). In the last quartile, which has the highest levels of informality, the relationship differs substantially from that observed in the first three quartiles. On average, markets in the last quartile of informality exhibit the lowest assortative matching: the average matching metric, weighted by employment, is 0.125, while for the first, second, and third quartiles the weighted averages are 0.248, 0.309, and 0.206, respectively. Furthermore, assortative matching does not appear to increase linearly with city size, as observed in the previous three quartiles; instead, the relationship remains flat, except for the largest markets.^{[22](#)}

²²The lower matching metric for markets with higher informality and the flat relationship between assortative match-

To address the possibility that this result is merely an artifact of benchmarking worker and firm quality against a national average or due to nonlinearities in the relationship between worker and firm effects, we recalculate the matching measure in Appendix Figures J.4, J.5, and J.9 using local Spearman ranks. These rank-based tests confirm that the negative sorting pattern persists: even relative only to other local options, high-performing workers in informal areas are not successfully pairing with high-performing firms.²³

We argue that informality itself acts as a structural friction through several mechanisms. For instance, our analysis, detailed in Appendix L, shows that higher local informality is associated with a significantly narrower supply of high-skill workers. This suggests that informality undermines returns to skill and discourages human capital accumulation, limiting the ability of formal firms to find suitably skilled matches. Moreover, as shown in Appendix K, the flat relationship between city size and matching quality remains evident even when analyzing large and small firms separately in high-informality markets.

Furthermore, we separate within-city matching from cross-city co-location patterns following Dauth et al. (2022). In Appendix M, we present an analysis confirming that mean worker and firm fixed effects are positively associated with population size and negatively associated with informality rates. We interpret this as suggesting that smaller, more informal LLMs host workers with lower fixed effects, on average. However, co-location does not explain the breakdown of within-city sorting in high-informality markets.

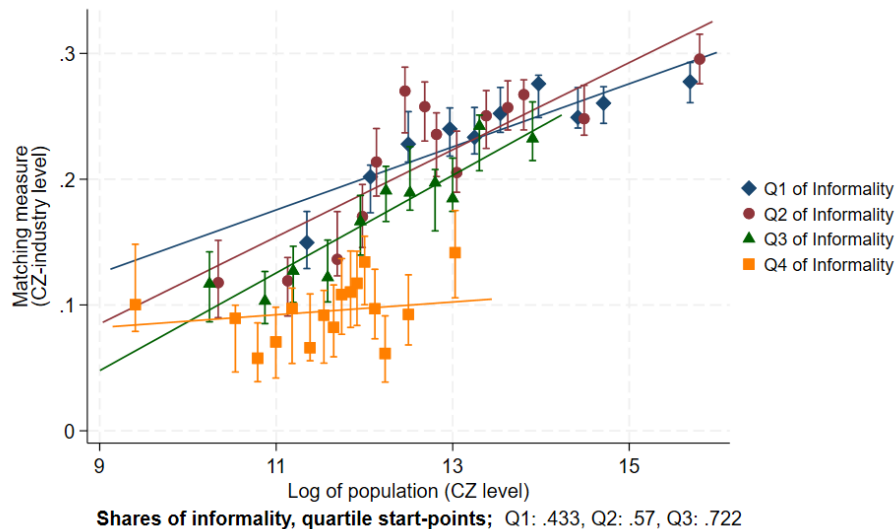
We also test whether the relationship holds for high- and low-mobility local labor markets by splitting the sample based on the mobility measure proposed by Jochmans and Weidner (2019). Appendix Figure J.6 shows that the reduced assortative matching in high informality areas holds even among low-mobility commuting-zone-sector cells, showing that the relationship in Figure 2 is not due to a higher proportion of low-mobility markets in the highest informality quartile.

ing and city size in these markets also appear at the commuting zone level (Appendix Figure J.1) and across periods (Appendix Figure J.2). We also observe a lower correlation among high-informality commuting zones if we exclude the southern region of Mexico from the estimation (Appendix Figure J.3).

²³Another possibility is that the distributions of worker and firm fixed effects differ across informality quartiles, for example, because there is a long tail of small, unproductive firms that tend to have informal workers to avoid paying social security contributions. If these firms also have workers with high fixed effects, it would be easier to find negative correlations between worker and firm fixed effects in the higher-informality quartiles, thereby depressing the overall relationship between labor market size and matching in the high-informality quartile. Appendix Table J.2 shows that this is not the case: across quartiles of informality, the variance of workplace effects is, if anything, lower in the higher informality quartile. We thank Gilles Duranton for pointing out this possibility.

Appendix Figure J.10 shows that the schooling results also hold in the separate samples.

Figure 2: City Size and Assortative Matching in Mexico’s Formal Labor Markets. Commuting Zone-Industry by Informality Quartiles



Source: Author’s calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching metric at the commuting zone-industry level for each quartile of the informality rate. Informality rates are assigned from the nearest available census: the average of 2000 and 2010 for 2004–2008, 2010 for 2009–2013, and 2015 for 2014–2018. Quartiles are calculated by pooling all years. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

We confirm this relationship by estimating the slope of the relationship between labor market size and matching separately by quartiles of labor informality in column (1) of Table 3. The slope for the first three quartiles of informality is positive and significant, while it is close to zero and precisely estimated for the markets with higher informality.²⁴

5.2 Schooling

Schooling levels may modify the magnitude of the assortative matching and the slope of its relationship with city size. Individuals with higher levels of education may be better informed about vacancies, the skill requirements of each job, and the productivity of particular firms. All of these

²⁴This result is further supported by Table J.1, which shows that even with more granular informality data at the CZ-sector level, the relationship between city size and sorting efficiency remains flat in the most informal local labor markets.

Table 3: City Size and Assortative Matching in Mexico's Formal Labor Markets: Controlling for Informality and Years of Schooling

Dependent variable: Matching metric - Correlation of worker and workplace FE				
	(1) Informality rate CZ-Industry	(2) Mean years of schooling CZ-Industry	Joint estimation	
			(3) Informality rate CZ-Industry	(4) Mean years of schooling CZ-Industry
Log Pop*first quartile	0.0250*** (0.004)	0.0208* (0.011)	0.0270*** (0.008)	
Log Pop*second quartile	0.0346*** (0.004)	0.0376*** (0.005)	0.0330*** (0.008)	0.0074 (0.008)
Log Pop*third quartile	0.0388*** (0.006)	0.0311*** (0.003)	0.0324*** (0.008)	0.0001 (0.008)
Log Pop*fourth quartile	0.0054 (0.011)	0.0211*** (0.003)	0.0016 (0.013)	-0.0081 (0.008)
Dummy first quartile	-0.1200** (0.058)	-0.1382 (0.135)	-0.1767* (0.104)	
Dummy second quartile	-0.2453*** (0.052)	-0.2865*** (0.061)	-0.2491*** (0.094)	-0.0617 (0.093)
Dummy third quartile	-0.3221*** (0.072)	-0.1882*** (0.047)	-0.2572** (0.101)	0.0365 (0.094)
Dummy fourth quartile	0.0126 (0.131)	-0.0246 (0.044)	0.0552 (0.146)	0.1585 (0.097)
Mean of dep. variable	0.180	0.180		0.180
R ²	0.455	0.451		0.457
N	10,096	10,096		10,096

Source: Author's calculations using IMSS and INEGI data. Columns (1) and (2) report regressions of our matching metric on log population interacted with quartiles of informality rates and quartiles of mean years of schooling, respectively, both measured at the commuting-zone level. Columns (3) and (4) under "joint estimation" report coefficients from a single regression that includes both interaction terms simultaneously, with column (3) displaying the coefficients on the informality interactions and column (4) displaying the coefficients on the years-of-schooling interactions. All regressions pool data from 2004-2008, 2009-2013, and 2014-2018 and include interval dummies. CZ stands for commuting zone. We restrict to cells with more than five firms and more than 50 workers. Clustered standard errors at the commuting-zone level in parentheses. *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

reasons may promote better matching for highly educated workers. As for the relationship with city size, part of the advantage of large cities in generating better matches between employers and employees may stem from the fact that more educated workers tend to move to larger cities, and firms in big cities demand more educated workers. If so, we would expect the relationship between city size and matching to be steeper among places where individuals are more educated.

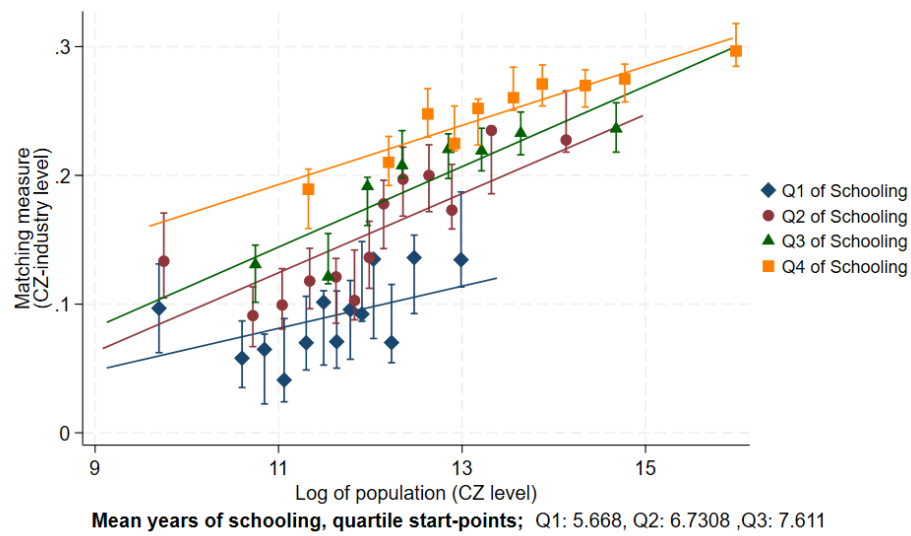
The evidence on the role of schooling in the city-size wage premium has been mixed. [De la Roca and Puga \(2016\)](#) do not find differences in the return to large-city experience between high- and low-education workers. However, they do observe differences in the returns to city size for high- and low-ability workers, measuring ability using worker fixed effects from a wage model. [Bacolod et al. \(2023\)](#) finds that high-learning-ability individuals sort to big cities. This sorting may influence both matching and education in larger cities. [Eckert et al. \(2022\)](#) show that the returns to big cities for migrants are larger for those with higher education, who find it easier to integrate into the labor markets of large cities.

Figure 3 shows that positive assortative matching is strongest in commuting zones with higher average schooling, and that the advantage of large cities in matching is also greater as average education rises in each labor market. For commuting zones in the top three schooling quartiles (with an average of more than 5.7 years of schooling), the matching metric is higher than in the bottom quartile, and it increases steeply with population, reaching nearly 0.3 in the largest commuting zones. By contrast, in low schooling markets (with an average of fewer than 5.7 years of schooling), the relationship is flatter.

This pattern suggests that higher levels of schooling are strongly correlated with positive assortative matching. More educated local labor markets may increase returns to skill complementarity and improve the efficiency of job search. These factors are associated with greater matching benefits and higher levels of matching quality. The evidence is consistent with the view that schooling is associated with higher productivity and with a market's tendency to generate productive matches.

The patterns in Figure 3 are confirmed in column (2) of Table 3, which shows a robust positive interaction between city size and average years of schooling on assortative matching. The estimated coefficients on log population in Table 3 are positive and significant across all schooling quartiles, with the most pronounced matching advantages appearing in the middle-to-high schooling brackets, even if the point estimates do not follow a strictly monotonic upward trend across all

Figure 3: City Size and Assortative Matching in Mexico's Formal Labor Markets. Commuting Zone-Industry by Years of Schooling Quartiles



Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and the estimated matching measure at the commuting zone-industry level for each quartile of the mean years of schooling. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands ([Cattaneo et al., 2025, 2024](#)) with default settings to generate the scatter plots.

four quartiles. Notably, the interaction effect for the bottom quartile is smaller and only marginally significant, consistent with a flatter, weaker relationship at low levels of schooling. This supports the interpretation that education is positively associated with the presence of matching externalities.²⁵

Finally, the joint results in columns (3) and (4) of Table 3 indicate that the correlation between schooling and matching is not independent of the local institutional environment. Column (3) reveals that the impact of informality remains essentially unchanged even after accounting for local schooling levels. Once informality rates are controlled for, the schooling interaction terms in column (4) become statistically insignificant and close to zero. The intuition is that schooling only enhances matching efficiency when the labor market is sufficiently formal to reward specialized skills. In high-informality environments, the returns to skill complementarity seem to be suppressed.

5.3 Firm Size and Firm Industry Composition

We now turn to differences in industry composition and firm size that distinguish Mexican firms from those elsewhere. Compared with Germany, where 71% of employment is in services (World Bank, 2023), Mexico has a larger share of employment in manufacturing. Furthermore, while Mexico is dominated by small firms—with approximately 93% of firms having five or fewer workers (Levy Algazi, 2018)—Germany’s firm-size distribution is more balanced toward medium and large enterprises.

We first examine whether matching externalities are stronger in markets with a larger share of workers in the service sector. Such a difference could naturally arise due to the higher job turnover that service industries tend to experience.²⁶ Figure 4a shows the relationship between city size and matching among service and non-service firms. The average matching measure outside the service sector is smaller, but the slopes of the relationship between matching and local labor market size

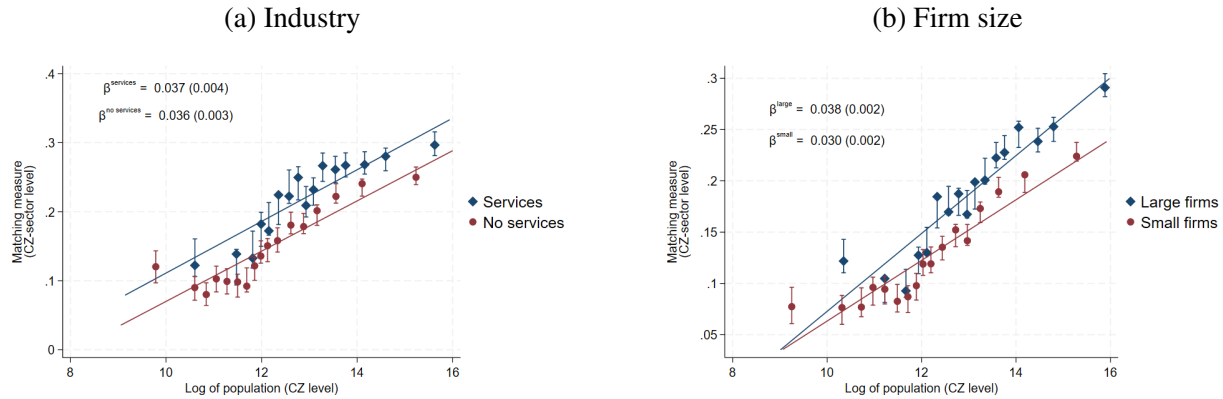
²⁵We examine this relationship at the commuting zone level in Appendix Figure J.7. At this level, positive assortative matching remains weaker in low-education areas. We also show estimates by time interval in Appendix Figure J.8. The analysis by period shows the same patterns, although the estimated binned scatter plots are noisier. Lastly, we ensure that these differences in matching externalities across informality quartiles are not driven by outliers: Figure J.11 shows similar results when excluding the largest and smallest areas.

²⁶Box 4 in Banco de México (2020) documents higher turnover in service industries relative to other industries in Mexico.

are similar.

Figure 4b estimates the strength of assortative matching and its relationship to labor market size by categorizing firms into two groups: large firms (with 16 or more workers) and small firms (with fewer than 16 workers).²⁷ Small firms exhibit weaker positive assortative matching than large firms on average. This is consistent with a search-costs channel: smaller firms may be unable to bear the costs of searching for better matches (Henderson, 1986; Helsley and Strange, 1990). Both small and large firms appear to benefit from larger city sizes by enhancing their ability to match with better workers, although the relationship is stronger for larger firms.

Figure 4: City Size and Assortative Matching in Mexico’s Formal Labor Markets: Industry and Firm Size Differences



Source: Author’s calculations using IMSS and INEGI data. Each panel displays a binned scatter plot of the log population and our matching metric at the commuting zone level. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. Panel (a) illustrates the relationship for service and non-service industries, respectively. We classify industries using 2-digit NAICS codes. Panel (b) shows the relationship separating large firms (16 or more workers) and small firms (fewer than 16 workers). The top-left values display the slope of the linear regression corresponding to the displayed relationship. The regression includes dummies for each time interval. Standard errors in parentheses. To produce the scatter plots, we used the binsreg command (Cattaneo et al., 2025, 2024) with the default settings. Clustered standard errors at the commuting-zone level in parentheses.

We further explore whether Mexico’s industry composition may lead to a different degree of matching externalities with city size through a counterfactual exercise, in which we estimate the relationship between city size and matching in Mexico if it had the same industrial composition

²⁷Levy Algazi (2018) classifies firms in Mexico by size into four size bins: 1-5 workers, 6-10, 10-50, and more than 50. Our classification allows for more firms in the “small firms” category.

as Germany. This exercise helps illustrate whether matching externalities are indeed similar in Mexico. For this analysis, we recalculate our matching measure at the commuting-zone level, reweighting individual observations to reflect each industry’s employment share in Germany.²⁸

Figure 5 shows the results of this exercise. Panel (a) illustrates the relationship between population and assortative matching at the commuting zone level in Mexico, restricting the sample to observations with available data on Germany’s industry composition. The estimated relationship is slightly steeper than that of Table 2, Panel A, column (1), using the whole sample. Panel (b) shows the relationship after reweighting the sample to match Germany’s industrial composition. The estimated relationship is 0.053, which is closer to the value estimated for Germany by Dauth et al. (2022). This larger slope suggests that Mexico’s economy has a bigger share of employment in industries where the association between matching and labor market size is weaker than in other industries. If Mexico were to have the same industrial composition as Germany, these matching externalities would be stronger. Besides a lower share of employment in services, Mexico also has higher employment shares in construction and mining than Germany, and these sectors tend to be less skill-intensive. The comparison between Germany and Mexico suggests that matching externalities are therefore lower in these sectors than in other parts of manufacturing and services.²⁹

²⁸We obtain data for Germany’s industrial composition for 2008-2018 from Eurostat. Then, we match industries at the NACE level to their NAICS counterparts to obtain German employment shares at the NAICS level. We exclude the first interval (2004-2008) of data because industrial composition data for Germany were unavailable. Furthermore, we are unable to match the following NAICS sectors to NACE data: educational services, health care and social assistance, other services (except public administration), and public administration.

We then obtain correlations between worker and firm effects at the commuting-zone level, reweighting to match Germany’s industrial composition. Note that we can obtain the correlation coefficients between worker and firm fixed effects at the commuting zone level through regression if we first standardize within each commuting zone and then estimate:

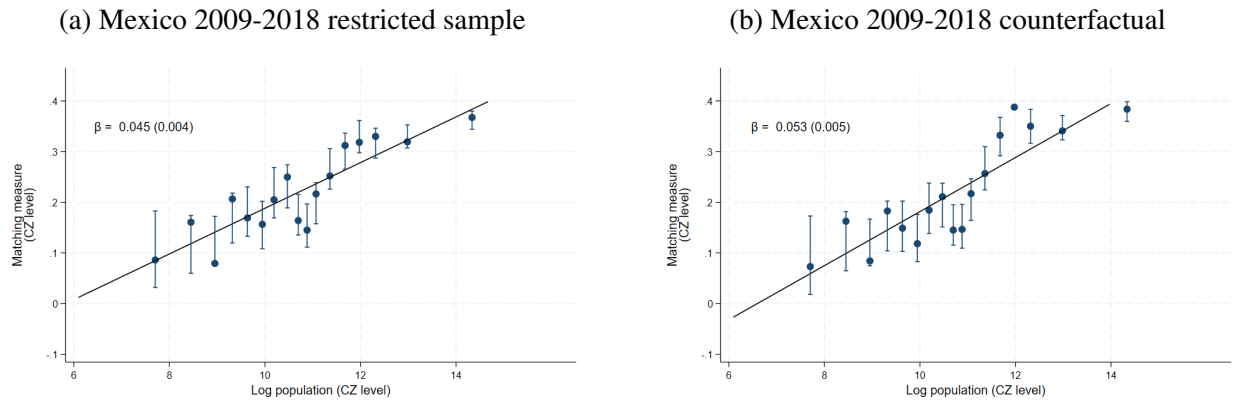
$$\text{standardized worker FE}_{ijt} = \alpha_{c(jt)} CZ_{c(jt)} + \rho_{c(jt)} CZ_{c(jt)} \text{standardized firm FE}_{jt} + \delta_t + \varepsilon_{ijt}.$$

In this equation, $CZ_{c(jt)}$ denotes an indicator for the commuting zone of firm j at time t . The coefficients $\rho_{c(jt)}$ are the estimated matching metrics at the commuting-zone level.

We then re-estimate this regression, weighting each observation by $w_{ij} = \theta_{s(j)t}^G / N_{s(j)}^{MX}$, the share of employment for sector $s(j)$ in Germany divided by the employment in that sector in Mexico. The coefficients from the reweighted regression serve as our industrial-composition-adjusted matching metrics.

²⁹We reproduce these results at the commuting zone level in Appendix Figure G.1 and Appendix Table G.1. The results at the commuting zone level are similar to those in the text. An additional concern is whether differences between Mexico and Germany stem from differences in commuting zone size. In Appendix Figure G.2 and Table G.2, we show that the results are similar when focusing on commuting zones with more than 50,000 inhabitants, which are more comparable to German commuting zones in size. In Appendix H, we conduct an additional robustness check that excludes government employees from the analysis. The results are similar to those reported in the main text. Not all public sector workers are included in the IMSS database because a separate institution, ISSSTE, manages their social security. According to official records, by December 2024, a total of 3,262,068 (73%) public-sector workers

Figure 5: City Size and Assortative Matching in Mexico's Formal Labor Markets: Germany's Industrial Composition Counterfactual.



Source: Author's calculations using IMSS, INEGI, and Eurostat data. Each panel displays a binned scatter plot of the log population and our matching measure at the commuting zone level. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. Panel (a) illustrates the relationship, restricting the sample to industries for which we have data on Germany's industrial composition. For Panel (b), we reweighted the sample to match Germany's industrial composition and re-estimated the correlation between worker and workplace fixed effects. To produce the scatter plots, we used the `binsreg` command (Cattaneo et al., 2025, 2024) with the default settings. Clustered standard errors at the commuting-zone level in parentheses.

6 Conclusion

We provide a descriptive analysis of Mexico’s underlying mechanisms of wage variation and worker-firm sorting patterns using social security records. We find a positive relationship between local labor market size and assortative matching between workers and firms. However, our findings highlight widespread informality as a structural feature associated with weaker matching gains in larger labor markets. Prevalent informality coincides with reduced positive assortative matching in the formal sector, a pattern consistent with weaker incentives to invest in specialization and firm-specific human capital. The result is that in larger cities, where a deeper labor market would typically strengthen worker-workplace matching, this advantage is attenuated by high informality.

In light of our findings, we believe there is a clear insight for labor market policy in Mexico. To strengthen matching externalities and more directly benefit from the productivity gains of labor market size, policymakers should prioritize improving local education and promoting greater formality in the labor market. In addition, policy designers must recognize that the success of any intervention aimed at improving the functioning of the labor market will depend heavily on the underlying level of informality in the local market; thus, general “blanket” interventions may have varying levels of success across regions.

Declaration of Generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work, the authors used ChatGPT 3.5, ChatGPT 4.0, and Grammarly to check spelling, improve clarity, and format BibTeX entries. After using these tools, the authors reviewed and edited the content as needed. They take full responsibility for the content of the publication.

were registered in the ISSSTE database, while 1,210,694 (27%) were registered in the IMSS database.

References

- Abowd, J. M., Kramarz, F., and Margolis, D. N. (1999). High Wage Workers and High Wage Firms. *Econometrica*, 67(2):251–333.
- Ahlfeldt, G. M. and Pietrostefani, E. (2019). The Economic Effects of Density: A Synthesis. *Journal of Urban Economics*, 111:93–107.
- Aldeco, L., Calderón, M., Chiquiar, D., Hanson, G., Pérez Pérez, J., and Velázquez, C. (2024). Local Labor Markets in Mexico: Definition, Databases, and Descriptive Analysis.
- Alix-Garcia, J. and Sellars, E. A. (2020). Locational Fundamentals, Trade, and the Changing Urban Landscape of Mexico. *Journal of Urban Economics*, 116:103213.
- Andersson, F., Burgess, S., and Lane, J. I. (2007). Cities, Matching and the Productivity Gains of Agglomeration. *Journal of Urban Economics*, 61:112–128.
- Bacolod, M., Blum, B. S., Rangel, M. A., and Strange, W. C. (2023). Learners in Cities: Agglomeration and the Spatial Division of Cognition. *Regional Science and Urban Economics*, 98:103838.
- Banco de México (2020). Informe Trimestral, Julio-Septiembre 2021.
- Banco de México (2023). Informe Trimestral, Julio-Septiembre 2023.
- Bassier, I. (2023). Firms and Inequality When Unemployment is High. *Journal of Development Economics*, 161:103029.
- Baum-Snow, N., Brandt, L., Henderson, J. V., Turner, M. A., and Zhang, Q. (2017). Roads, Railroads, and Decentralization of Chinese Cities. *The Review of Economics and Statistics*, 99(3):435–448.
- Baum-Snow, N. and Pavan, R. (2012). Understanding the City Size Wage Gap. *The Review of Economic Studies*, 79(1):88–127.
- Behrens, K., Duranton, G., and Robert-Nicoud, F. (2014). Productive Cities: Sorting, Selection, and Agglomeration. *Journal of Political Economy*, 122(3):507–553.
- Bleakley, H. and Lin, J. (2012). Thick-Market Effects and Churning in the Labor Market: Evidence from US Cities. *Journal of Urban Economics*, 72(2-3):87–103.
- Bonhomme, S., Lamadon, T., and Manresa, E. (2019). A Distributional Framework for Matched Employer Employee Data. *Econometrica*, 87(3):699–739.
- Cadena, B. (2014). Recent Immigrants as Labor Market Arbitrageurs: Evidence from the Minimum Wage. *Journal of Urban Economics*, 80(C):1–12.
- Card, D., Cardoso, A. R., Heining, J., and Kline, P. (2018). Firms and Labor Market Inequality: Evidence and Some Theory. *Journal of Labor Economics*, 36(S1):S13–S70.

- Card, D., Heining, J., and Kline, P. (2013). Workplace Heterogeneity and the Rise of West German Wage Inequality. *The Quarterly Journal of Economics*, 128(3):967–1015.
- Cattaneo, M. D., Crump, R. K., Farrell, M. H., and Feng, Y. (2024). On Binscatter. *American Economic Review*, 114(5):1488–1514.
- Cattaneo, M. D., Crump, R. K., Farrell, M. H., and Feng, Y. (2025). Binscatter Regressions. *Stata Journal*, 25(1):3–50.
- Chauvin, J. P., Glaeser, E., Ma, Y., and Tobio, K. (2017). What is Different About Urbanization in Rich and Poor Countries? Cities in Brazil, China, India and the United States. *Journal of Urban Economics*, 98:17–49.
- Combes, P.-P., Démurger, S., Li, S., and Wang, J. (2020). Unequal Migration and Urbanisation Gains in China. *Journal of Development Economics*, 142:102328.
- Dauth, W., Findeisen, S., Moretti, E., and Suedekum, J. (2022). Matching in Cities. *Journal of the European Economic Association*, 20(4):1478–1521.
- D’Costa, S. and Overman, H. G. (2014). The Urban Wage Growth Premium: Sorting or Learning? *Regional Science and Urban Economics*, 48:168–179.
- De la Roca, J., Parkhomenko, A., and Velásquez-Cabrera, D. (2023). Skill Allocation and Urban Amenities in the Developing World. Working paper.
- De la Roca, J. and Puga, D. (2016). Learning by Working in Big Cities. *The Review of Economic Studies*, 84(1):106–142.
- Diallo, Y., Sarr, I., and Diagne, I. (2022). Role of Firms in Wage Dispersion: Evidence from a Developing Country. *PEDL Research Papers*.
- Duranton, G. (2015). Growing Through Cities in Developing Countries. *World Bank Research Observer*, 30(1):39–73.
- Duranton, G. (2016). Agglomeration Effects in Colombia. *Journal of Regional Science*, 56(2):210–238.
- Duranton, G. and Puga, D. (2004). Micro-Foundations of Urban Agglomeration Economies. In *Handbook of Regional and Urban Economics*, volume 4, pages 2063–2117. Elsevier.
- Eckert, F., Hejlesen, M., and Walsh, C. (2022). The Return to Big-City Experience: Evidence from Refugees in Denmark. *Journal of Urban Economics*, 130:103454.
- Fowler, C. S. and Jensen, L. (2020). Bridging the Gap Between Geographic Concept and the Data We Have: The Case of Labor Markets in the USA. *Environment and Planning A: Economy and Space*, 52(7):1395–1414.
- Frías, J. A., Kaplan, D. S., Verhoogen, E., and Alfaro-Serrano, D. (2022). Exports and Wage Premia: Evidence from Mexican Employer-Employee Data. *The Review of Economics and Statistics*, pages 1–45.

- Ghani, E., Goswami, A. G., and Kerr, W. R. (2016). Highway to Success: The Impact of the Golden Quadrilateral Project for the Location and Performance of Indian Manufacturing. *The Economic Journal*, 126(591):317–357.
- Gould, E. D. (2007). Cities, Workers, and Wages: A Structural Analysis of the Urban Wage Premium. *The Review of Economic Studies*, 74(2):477–506.
- Grover, A., Lall, S., and Timmis, J. (2023). Agglomeration Economies in Developing Countries: A Meta-Analysis. *Regional Science and Urban Economics*, 101:103901.
- Helsley, R. W. and Strange, W. C. (1990). Matching and Agglomeration Economies in a System of Cities. *Regional Science and Urban Economics*, 20(2):189–212.
- Henderson, J. V. (1986). Efficiency of Resource Usage and City Size. *Journal of Urban Economics*, 19(1):47–70.
- Jedwab, R., Ianchovichina, E., and Haslop, F. (2022). Consumption Cities versus Production Cities: New Considerations and Evidence. Policy Research Working Paper WPS 10105, World Bank Group, Washington, D.C.
- Jochmans, K. and Weidner, M. (2019). Fixed-Effect Regressions on Network Data. *Econometrica*, 87(5):1543–1560.
- Kline, P., Saggio, R., and Sølvesten, M. (2020). Leave-out Estimation of Variance Components. *Econometrica*, 88(5):1859–1898.
- Kumler, T., Verhoogen, E., and Frías, J. (2020). Enlisting Employees in Improving Payroll Tax Compliance: Evidence from Mexico. *Review of Economics and Statistics*, 102(5):881–896.
- Levy Algazi, S. (2018). *Under-rewarded Efforts: The Elusive Quest for Prosperity in Mexico*. Inter-American Development Bank.
- Millimet, D. L. and Bellemare, M. F. (2025). On the (Mis)Use of the Fixed Effects Estimator. *Oxford Bulletin of Economics and Statistics*.
- Moretti, E. and Yi, M. (2024). Size Matters: Matching Externalities and the Advantages of Large Labor Markets. Working Paper 32250, National Bureau of Economic Research.
- Pérez Pérez, J. and Nuño-Ledesma, J. G. (2024). Workers, Workplaces, Sorting, and Wage Dispersion in Mexico. *Economía LACEA Journal*, 23(1).
- Petrongolo, B. and Pissarides, C. (2006). Scale Effects in Markets with Search. *The Economic Journal*, 116(508):21–44.
- Puggioni, D., Calderón, M., Cebaleros Zurita, A., Fernández Bujanda, L., Inguanzo González, J. A., and Jaume, D. (2022). Inequality, Income Dynamics, and Worker Transitions: The case of Mexico. *Quantitative Economics*, 13(4):1669–1705.

- Rotemberg, J. J. and Saloner, G. (2000). Competition and Human Capital Accumulation: A Theory of Interregional Specialization and Trade. *Regional Science and Urban Economics*, 30(4):373–404.
- Ulyssea, G. (2010). Regulation of Entry, Labor Market Institutions and the Informal Sector. *Journal of Development Economics*, 91(1):87–99.
- Ulyssea, G. (2018). Firms, Informality, and Development: Theory and Evidence from Brazil. *American Economic Review*, 108(8):2015–2047.
- Wheeler, C. H. (2008). Local Market Scale and the Pattern of Job Changes Among Young Men. *Regional Science and Urban Economics*, 38(2):101–118.
- World Bank (2023). Employment in Services (% of Total Employment). Accessed: 2023-06-10.

Online Appendix - Not for Publication

A Data and Descriptive Statistics

We use IMSS data previously analyzed in [Pérez Pérez and Nuño-Ledesma \(2024\)](#). IMSS refers to the Instituto Mexicano del Seguro Social, a Mexican government agency responsible for public health, pension management, and social security. Private sector employees are required by law to register with IMSS. According to Mexico’s labor survey, about 83% of the formal workforce in 2022 was registered with IMSS. Self-employed individuals may register with IMSS, granting them access to certain aspects of the social security system. Typically, self-employed workers register with the equivalent of one legal minimum salary. Self-employed records account for approximately 0.1% of the overall IMSS database. In cases where a worker reports multiple jobs, we retain the job with the highest reported wage. Only 2.5% of workers reported having more than one job in December 2018.

The IMSS social security information is available every month. We analyze records spanning from November 2004 to December 2018. Our analysis concludes in 2018 due to significant changes in Mexico’s labor market resulting from the COVID-19 pandemic and substantial increases in the minimum wage from 2019 to 2022. The database initially consisted of 12.8 million workers in November 2004 and grew to 20.1 million workers by December 2018. Our primary variable of interest is daily taxable income, which encompasses various forms of compensation and excludes additional non-taxable payments, such as paid vacations and bonuses. Wages exceeding 25 UMAs (units of measure and update) are capped. About 1,539 pesos per day, or approximately 80 USD at 2018 exchange rates. We rely on the *registro patronal* as our workplace identifier, rather than alternatives such as the *Registro Federal de Contribuyentes* (RFC), because *registros* better capture employer-worker relationships in settings where multiple employers operate within the same plant. The RFC is an identification number issued by the Mexican Tax Administration Service to individuals, firms, and any other legal entities for tax compliance purposes. The *registro patronal* is an employer registry identification number assigned to all employers to manage pensions and provide other social security benefits through the Mexican Social Security Institute. The *registro patronal* identifiers are anonymized by our data provider before access is granted. The procedure used to mask the identifiers is inconsequential to our analysis and does not affect our

econometric analysis. Because the *registro patronal* is employer-based rather than plant-based, our results reflect employer effects rather than the plant effects traditionally reported in the literature using AKM models (e.g., [Card et al. 2013](#)).

Table A.1, reproduced from [Pérez Pérez and Nuño-Ledesma \(2024\)](#), summarizes the IMSS data. We divide our data into three time intervals: 2004-2008, 2009-2013, and 2014-2018. Within each year, our sample includes a substantial number of wage observations, ranging from 73 to 113 million for men aged 25-54. In column (2) of the table, we observe a 0.7% decrease in the average real daily wage for prime-age men between 2009 and 2014 when compared to 2005. However, this decline is followed by a 1.5% increase by 2018. Column (3) illustrates a widening gap in earned wages between 2005 and 2018.

Table A.1: Descriptive Statistics: Prime-age Men, National Level

	Real wage			
	(1)	(2)	(3)	(4)
	Observations	Mean	Std. dev	Percent censored
2005	73,847,545	394.589	406.167	2.675
2009	80,065,916	394.602	402.992	2.690
2014	96,354,574	394.200	409.212	2.649
2018	110,844,774	401.186	412.367	2.058

Source: Authors' calculations using IMSS data. Observations correspond to the sum of all the monthly observations in a year. Real wages are daily taxable income registered with the IMSS, expressed in real terms from July 2018. The percent censored is the percentage of observations with wages exactly equal to the upper wage limit of 25 minimum wages or units of measure, which are updated per year.

[Abowd et al. \(1999\)](#) show that AKM estimates identify fixed effects for workers and workplaces within a "connected set" of workplaces that share a pool of workers who switch jobs at least once. Our estimates use the largest connected set within each time interval. A workplace is part of the connected set if at least one of its workers has worked or will work at another workplace during the given time interval. Direct connections between every pair of workplaces are not necessary for a connected set to exist.

Table A.2, reproduced from [Pérez Pérez and Nuño-Ledesma \(2024\)](#), shows the number of worker-month observations for prime-age men with more than one job, the number of individuals, and the average and standard deviation of log wages. In each interval, our database comprises

324 to 518 million worker-month observations, representing 12 to 16 million individuals. The standard deviation of salaries slightly increased from 0.81 in the 2004-2008 interval to 0.83 in the 2014-2018 interval. Average real wages exhibited a consistent upward trend throughout the sample. Columns (5) to (8) of Table [A.2](#) display the corresponding descriptive statistics for the largest connected set of prime-age male workers. The largest connected set encompasses at least 96% of all worker-year observations and 96% of all individuals within a given interval. Average wages within the connected set are slightly higher than those in the overall sample, while standard deviations are marginally smaller. Given the substantial size of the connected set relative to the entire sample, the similar mean salaries, standard deviations, and comparable trends in average wages and salary dispersion, our focus on this connected group does not involve a significant loss of detail.

Table A.2: Descriptive Statistics - Overall Sample and Workers in the Largest Connected Set

Interval	All sample				Individuals in largest connected set			
	(1) All obs.	(2) Individuals	Log wage		(5) All obs.	(6) Individuals	Log wage	
			(3) Mean	(4) Std. dev.			(7) Mean	(8) Std. dev.
Nov 2004-2008	324,468,447	11,835,313	5.627	0.813	311,941,032	11,363,073	5.657	0.808
Ratio: largest connected/all					96.14	96.01	100.53	99.39
2009-2013	431,227,399	13,526,466	5.600	0.826	417,008,147	13,083,589	5.625	0.823
Ratio: largest connected/all					96.70	96.73	100.45	99.65
2014-2018	518,128,252	15,920,775	5.609	0.831	505,015,793	15,512,438	5.628	0.829
Ratio: largest connected/all					97.47	97.44	100.35	99.71
Change from first to last interval			-0.018	0.018			-0.029	0.021

Source: Authors' calculations using IMSS data. Statistics for men aged 25 to 54 years who held more than one job, i.e., were employed in more than one workplace during the analysis period. Log wage is the log of daily taxable income registered with IMSS, expressed in real terms using July 2018 prices. "Ratio: largest connected/all" is the ratio of the corresponding statistic in the largest connected set to its counterpart in the full sample.

We complement the IMSS data with LLM characteristics. At the LLM level, we calculate the population using municipality-level data from Mexican censuses and intercensal surveys. We also obtain the informality rate and the mean years of schooling at the LLM level from [Aldeco et al. \(2024\)](#). At the city level, we calculate unionization rates using the Occupation and Employment Survey of the National Institute of Statistics and Geography (INEGI). The informality rate is the percentage of workers without social security benefits over the total number of workers. The unionization rate is the percentage of formal workers who report belonging to a union, divided by the total number of formal workers.

B Imputation of censored wages

According to the IMSS database documentation, daily earnings were censored at 25 daily minimum wages before February 2017 and at 25 "units of measurement and actualization" (UMA) thereafter. On average, these values amount to 1,539 pesos per day, which corresponds to approximately 80 USD at 2018 exchange rates. The censoring may affect our estimates of wage variance and the fraction of that variance attributable to positive assortative matching. Therefore, for our main analysis, we impute wage values for these top-coded observations.

While wages should be top-coded at either 25 minimum wages or 25 UMAs, in practice, we find some wage values exceeding these censoring limits. Therefore, to impute daily earnings for top-coded observations, we examined the right tail of the wage distribution starting at the documented censoring threshold. Specifically, we plotted the wage density and searched for the most frequent value immediately above the theoretical censoring point. We interpret this modal spike as the actual censoring point. This empirical approach reflects the idea that, under censoring, a large number of observations will accumulate at the top-coding threshold. To account for potential reporting errors or rounding, we allowed a 5-peso margin above the documented value when identifying the mode.

After identifying the censoring thresholds, we followed the methodology in [Card et al. \(2013\)](#) to impute censored wages. Specifically, we estimate Tobit models of wages, regressing them on a set of independent variables and obtaining predictions for the wages of top-coded observations. The independent variables in each model include the worker's age; the number of employees in

the firm; a dummy indicating whether the firm has 10 or more employees; the average wage in the firm; the fraction of employees in the firm with censored wages; the fraction of times the worker's wage is censored in the sample; and a dummy indicating if the worker is observed in only one month.

Estimating the Tobit model in our individual-level data proved unfeasible. We note, however, that the estimates of the Tobit model coefficients, except for the residual variance, can be obtained by estimating on a cell-aggregated dataset using frequency weights. Therefore, we estimate two Tobit regressions: a first one using a 1% random sample of workers, including the previously mentioned independent variables, and a second one aggregating the data by period, region, industry, workers observed only once in the database, and age group.³⁰

We estimate over 169 Tobit regressions using aggregated data, one for each month from November 2004 to December 2018.³¹ The separate estimation allows model coefficients and censoring thresholds to vary over time, ensuring that predicted values reflect time-specific wage structures. With the estimated coefficients from the second Tobit model and the variance of the residuals retrieved from the worker-level Tobit model in the 1% sample, we apply the formula proposed by [Card et al. \(2013\)](#) to estimate the uncensored log wage y'' for each censored observation:

$$y'' = X'\hat{\beta} + \hat{\sigma}\Phi^{-1}[k + u \times (1 - k)]. \quad (\text{B.1})$$

Here, the variable y'' is the imputed uncensored log wage, $X'\hat{\beta}$ is the predicted mean from the Tobit model, $\hat{\sigma}$ is the estimated standard deviation of the residuals of the Tobit model, the function Φ^{-1} is the inverse CDF of the standard normal distribution, and the variable $k = \Phi\left(\frac{c - X'\hat{\beta}}{\hat{\sigma}}\right)$ represents the CDF evaluated at the standardized censoring point c . Additionally, the variable u is a random draw from a uniform distribution. This approach generates plausible values for censored wages by leveraging the Tobit model's distributional assumptions and introducing randomness through u , ensuring variability in the imputed values.

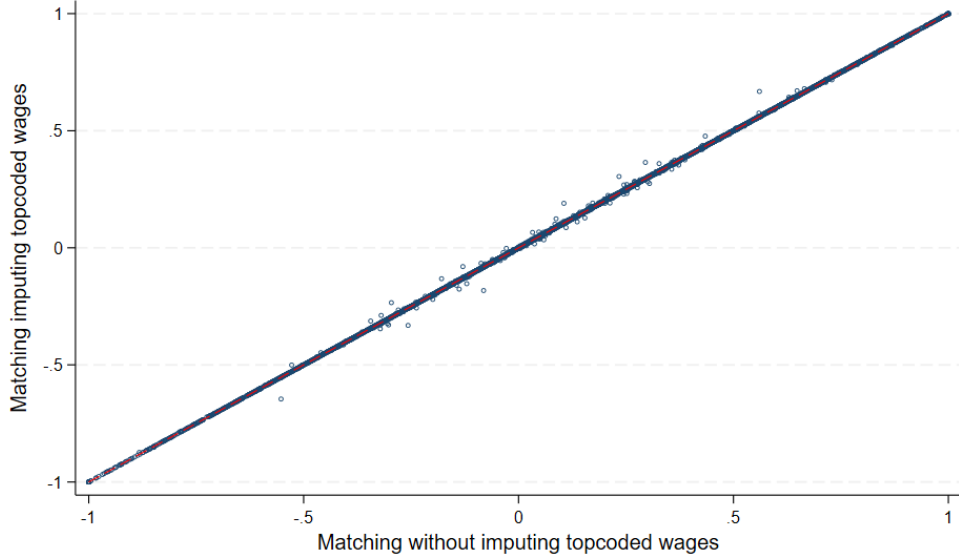
After imputing wages for the censored workers, we construct our matching metric using data

³⁰This approach follows the methodology proposed by [Card et al. \(2013\)](#), who estimate separate Tobit models on cells defined by individual and firm characteristics. We are unable to fully replicate their cell construction since we lack information on some firm-level educational characteristics, such as the mean years of schooling and the fraction of university graduates at the current firm.

³¹We follow [Card et al. \(2013\)](#), who estimate a series of Tobit models by year.

that include the imputed wages. We then compare the matching metric before and after the imputation. Figure B.1 shows that correlations at the commuting zone-industry level remained stable, suggesting that the imputation does not meaningfully alter our estimates.

Figure B.1: Comparison of Correlations with and without Imputation of Censored Daily Wages



Source: Authors' calculations using IMSS data.

C Details about Commuting Zone Construction

We use commuting zones for Mexico calculated by [Aldeco et al. \(2024\)](#). We provide a summary of their methodology for building commuting zones here. Using residence-workplace data from the 2010 Mexican census, [Aldeco et al. \(2024\)](#) groups municipalities by their similarity in commuting patterns, using the same methodology as [Fowler and Jensen \(2020\)](#) for the US. First, they calculate a commuting dissimilarity index for each pair of municipalities i, j , using data on commuting flows f_{ij} from an origin-destination survey:

$$D_{ij} = 1 - \frac{f_{ij} + f_{ji}}{\min(\sum_l f_{il}, \sum_l f_{lj})} \quad (\text{C.2})$$

This index increases as the share of workers commuting between municipalities i and j decreases. After building the index, they cluster the municipalities based on this index using a hier-

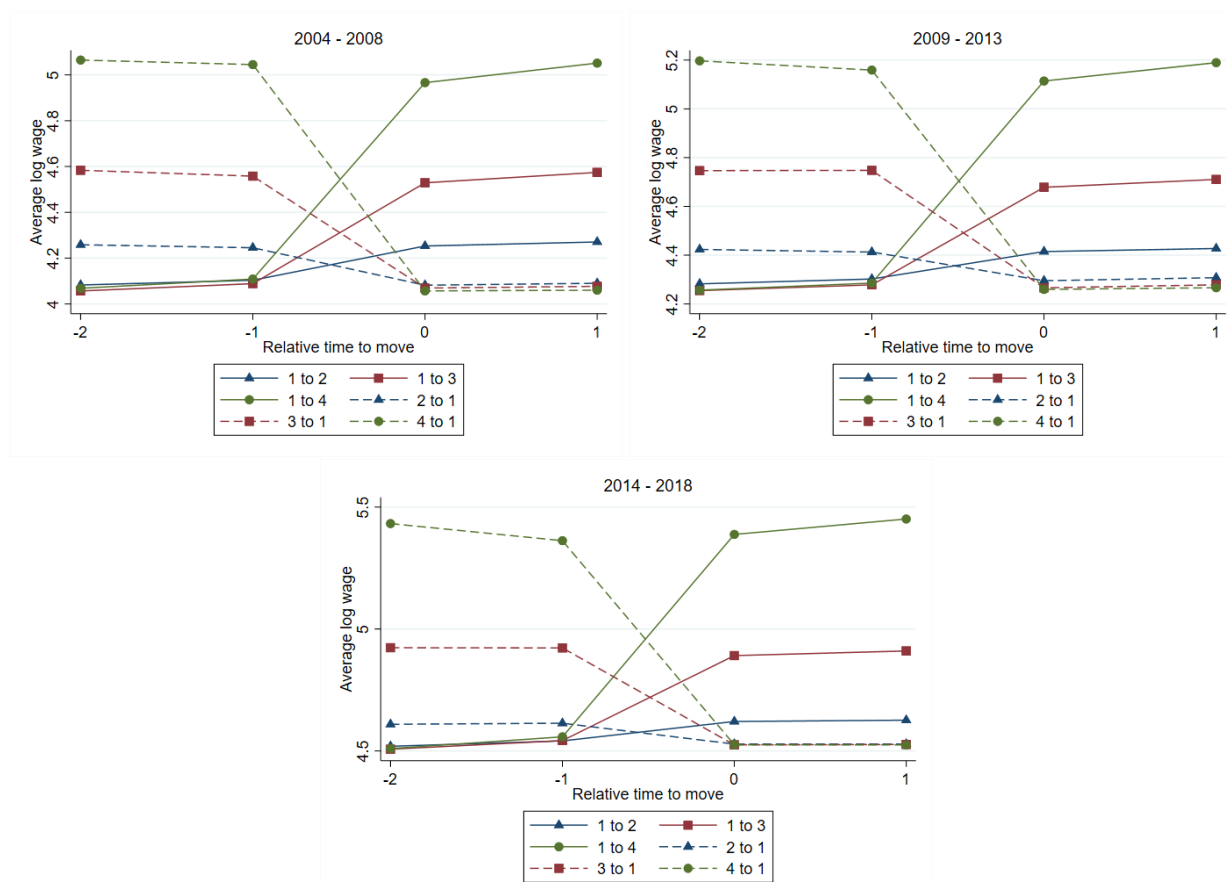
archical clustering algorithm.

D Exchangeability

In this section, we reproduce the evidence for exchangeability shown in [Pérez Pérez and Nuño-Ledesma \(2024\)](#). According to [Card et al. \(2013\)](#), if the residual term in equation (1) is uncorrelated with the variables on the right-hand side, workers who move from workplace A to workplace B should, on average, experience a wage change opposite in sign to workers moving in the opposite direction. To explore this in our dataset, we follow [Card et al. \(2013\)](#) and present an event study in Figure D.1, adapted from [Pérez Pérez and Nuño-Ledesma \(2024\)](#). The plot shows the average wages of workers who changed jobs during each interval in our analysis period. These workers may transition from "low-wage" to "high-wage" workplaces or vice versa. We categorize workplaces based on the quartiles of the average wages of their co-workers in the initial and final jobs. We then calculate average wages before and after the job switch for each category. Our analysis excludes observations from establishments with only one worker and focuses on "direct" moves, which are defined as moves without an unemployment spell between jobs.

Figure D.1 reveals that different mobility groups, classified by the average wage of co-workers, have distinct average wage levels before and after a job move. Before the move, average wages in the quartile of origin vary monotonically with the destination quartile. For instance, workers moving from quartile 4 (the highest average co-worker wage) to quartile 1 (the lowest average co-worker wage) have higher average wages before the job switch than those moving from quartile 3 to quartile 1, and so on. Additionally, the absolute change in average salaries when transitioning from one quartile to another is equivalent in magnitude to the variation associated with the opposite change. This symmetry aligns with an additive wage model that incorporates worker and workplace fixed effects, similar to the one we estimate.

Figure D.1: Exchangeability: Average Log Wage Around Movement by Quartile of Average co-workers' Wages in the Origin and Destination Workplace



Source: Authors' calculations using IMSS data. The graph shows the average wages of workers who move between an origin and a destination workplace, from two months before the move to one month after. The lines group workers by quartiles of average co-worker wages at the origin and destination workplaces. The panels correspond to different time intervals. We exclude observations from establishments with only one worker. We retain only "direct" moves that do not involve an unemployment spell between jobs.

E Additional Estimates

Table E.1: Estimates with Log Employment as Independent Variable

	(1) CZ	(2) CZ-Industry
Dependent variable: Matching metric - Correlation of worker and workplace FE		
Log employment	0.0279*** (0.002)	0.0325*** (0.002)
R ²	0.113	0.059
N	1,911	10,097
Mean of dep. variable	0.196	0.180

Source: Author's calculations using IMSS data. The columns display the results of regressions of the correlation coefficient between worker and workplace effects from AKM model estimates and log employment, at the commuting zone and commuting zone industry levels. All the regressions pool data across the three intervals: 2004-2008, 2009-2013, and 2014-2018, and include interval dummies. CZ stands for commuting zone. For column (2), we restrict to cells with more than 50 workers and more than 5 firms. Clustered standard errors at the commuting-zone level in parentheses. *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table E.2: City Size and Assortative Matching in Mexico's Formal Labor Markets: Estimates by Interval

Dependent variable: Matching metric - Correlation of worker and workplace FE		
	(1) CZ	(2) CZ-Industry
A: 2004-2008		
Log Population	0.0355*** (0.007)	0.0458*** (0.003)
N	596	3,200
R ²	0.051	0.072
Mean of dep. variable	0.177	0.159
B: 2009-2013		
Log Population	0.0426*** (0.006)	0.0410*** (0.003)
N	653	3,381
R ²	0.084	0.061
Mean of dep. variable	0.188	0.175
C: 2014-2018		
Log Population	0.0428*** (0.005)	0.0368*** (0.003)
N	664	3,516
R ²	0.096	0.059
Mean of dep. variable	0.220	1.000

Source: Author's calculations using IMSS and INEGI data. The columns display the results of regressions of our matching measure and log employment, at the commuting zone and commuting zone industry levels. Each Panel corresponds to a different time interval. CZ stands for commuting zone. For column (2), we restrict to cells with more than 50 workers and more than five firms. Clustered standard errors at the commuting-zone level in parentheses. *: p<0.1, **: p<0.05, ***: p<0.01.

Table E.3: City Size and Assortive Matching in Mexico's Formal Labor Markets: Heteroskedasticity adjustment.

Dependent variable: Matching metric - Correlation of worker and workplace FE		
	(1)	(2)
	CZ	CZ-Industry
A: Baseline Model		
Log Population	0.0408*** (0.004)	0.0410*** (0.003)
B: Weighted Least Squares (non-parametric weights)		
Log Population	0.0381*** (0.002)	0.0377*** (0.001)
N	1,913	10,097
Mean of dep. variable	0.196	0.180

Source: Author's calculations using IMSS and INEGI data. This table addresses heteroskedasticity concerns through WLS following the agnostic approach in [Cadena \(2014\)](#). Specifically, we estimate the baseline model and then regress the squared residuals of this baseline regression on cell size with a local linear regression model. We then re-estimate the baseline regression, weighting each observation by the inverse of the variance conditional on cell size as predicted by the local linear regression model. We iterate this process three times. We use an Epanechnikov kernel for the local-linear regression with the bandwidth determined by minimizing the MSE of the prediction. Clustered standard errors at the commuting-zone level in parentheses. *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

F Union Coverage

The study of unionization as a potential limiting factor for positive assortative matching in labor markets dates back to [Card et al. \(2013\)](#), who attribute the increased relevance of assortative matching to wage inequality in Germany to a decline in union power. Unions may prevent the rotation and dismissal of union-affiliated workers, limiting assortative matching.

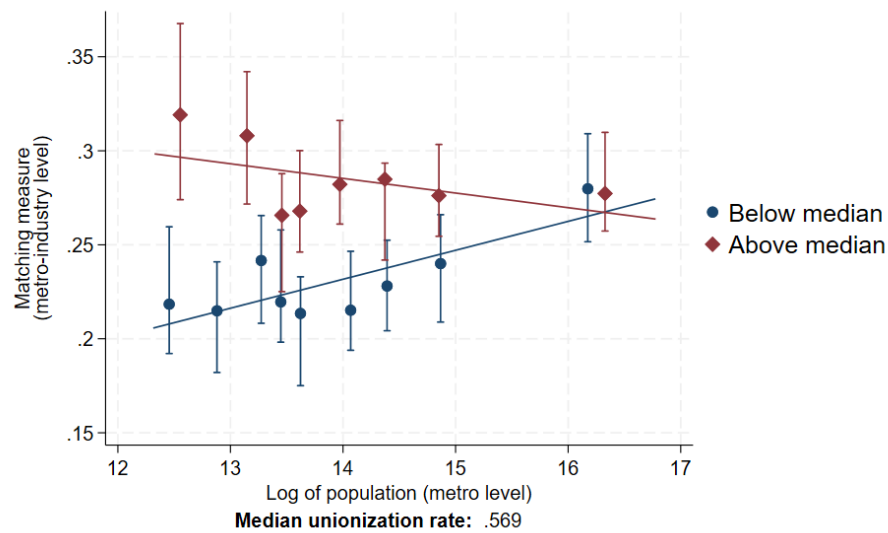
Due to data limitations, unionization rates are only available at the metro-industry level, rather than at our preferred geographical unit of analysis (CZ and CZ-industry level). As a result, these results are not directly comparable to those obtained for mechanisms such as schooling and informality. Nevertheless, they offer complementary evidence on the role of institutional factors—such as union presence—in shaping assortative matching dynamics across local labor markets.

We explore the relationship between unionization rates and assortative matching in local labor markets in Figure [F.1](#). We split the sample at the median unionization rate (56.9%). Our findings are mixed. In areas with below-median unionization, the strength of positive assortative matching is lower than in areas with above-median unionization. On the other hand, in metropolitan areas with below-median unionization, we observe a positive relationship between matching and city size: assortative matching improves as local labor market size increases. In contrast, for metropolitan areas with above-median unionization rates, this relationship becomes less pronounced.

This pattern suggests two possible interpretations. First, strong unions may institutionalize wage and employment norms that substitute for market-based matching, leading to hiring practices less dependent on productivity alignment. Second, relatively rigid labor institutions may constrain the flexibility needed to exploit agglomeration effects, muting the advantages of urban scale.

These results are consistent with the idea that strong unions flatten the relationship between city size and sorting, either by enforcing non-market job-allocation rules or by limiting firm and worker discretion in hiring decisions.

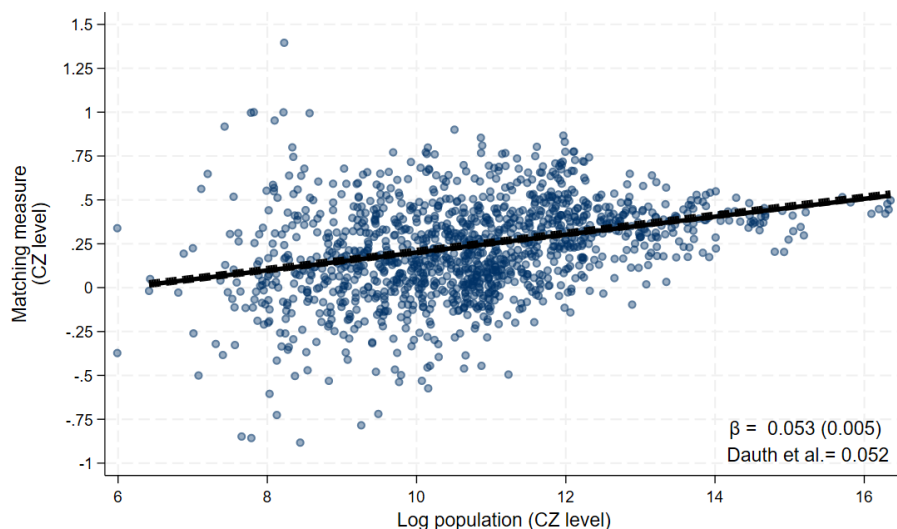
Figure F.1: City Size and Assortative Matching in Mexico's Formal Labor Markets. Metropolitan Area-Industry and Unionization Rate



Source: Author's calculations using IMSS and INEGI data. The figure shows a binned scatter plot of the log population and the correlation between estimated worker and workplace effects from AKM models at the metro-industry level for units above and below the median unionization rate. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

G Counterfactual Estimates with Germany's Industrial Composition and City Size Distribution

Figure G.1: City Size and Assortative Matching in Mexico's Formal Labor Markets. Germany's Industrial Composition Counterfactual at the Commuting Zone Level.



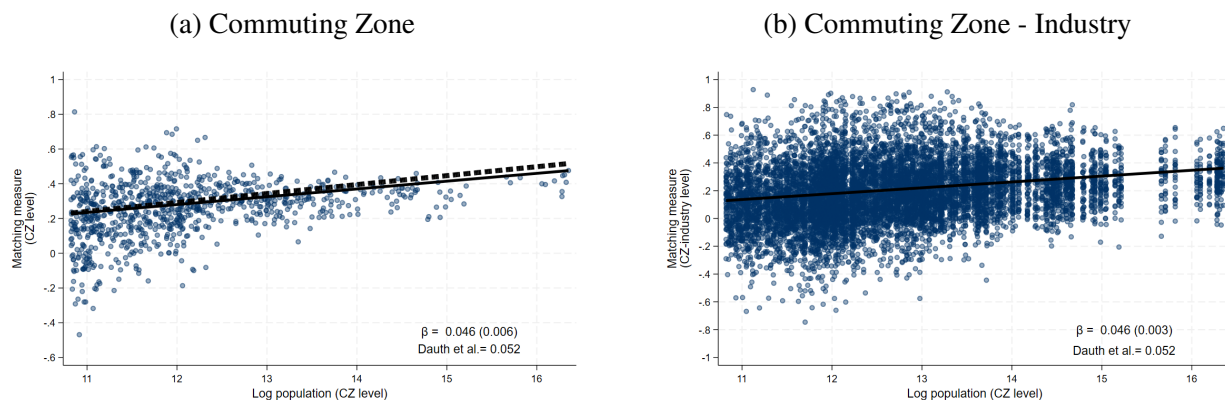
Source: Author's calculations using IMSS, INEGI, and Eurostat data. The figure displays a scatterplot illustrating the relationship between log population and our matching metric at the commuting zone level. We reweighted the sample to match Germany's industrial composition and re-estimated the correlation between worker and workplace fixed effects. For comparison, the figure displays the fitted line with the slope estimated for Germany by [Dauth et al. \(2022\)](#) corrected for limited-mobility bias. The bottom-right values indicate the slope of the linear regression for the displayed relationship. The regression includes dummies for each time interval. Clustered standard errors at the commuting zone level in parentheses.

Table G.1: City Size and Assortative Matching in Mexico’s Formal Labor Markets. Germany’s Industrial Composition Counterfactual.

Dependent variable: Matching metric - Correlation of worker and workplace FE	
	(1)
	CZ
Germany’s industrial composition counterfactual	
Log Population	0.0534*** (0.005)
R ²	0.111
N	1,300
Mean of dep. variable	0.219

Source: Author’s calculations using IMSS, INEGI, and Eurostat data. “CZ” stands for commuting zone. Regressions pool data from 2009-2013 and 2014-2018 with interval dummies. We reweighted the sample to match Germany’s industrial composition and re-estimated the correlation between worker and workplace fixed effects. Clustered standard errors at the commuting-zone level in parentheses. *: p<0.1, **: p<0.05, ***: p<0.01.

Figure G.2: City Size and Assortative Matching in Mexico’s Formal Labor Markets. Excluding Commuting Zones with less than 50,000 Inhabitants.



Source: Author’s calculations using IMSS and INEGI data. Each Panel displays a scatterplot illustrating the relationship between log population and our matching metric at the commuting zone and commuting zone industry levels. We exclude CZs with fewer than 50 inhabitants. For comparison, Panel (a) displays the fitted line with the slope estimated for Germany by [Dauth et al. \(2022\)](#) corrected for limited-mobility bias. For Panel (b), we restrict to cells with more than five firms and more than 50 workers. We classify industries using 2-digit NAICS codes. The bottom-right values show the slope of the linear regression for the displayed relationship. The regressions include dummies for each time interval. Clustered standard errors at the commuting-zone level in parentheses.

Table G.2: City Size and Assortative Matching in Mexico's Formal Labor Markets. Excluding Commuting Zones with less than 50,000 Inhabitants.

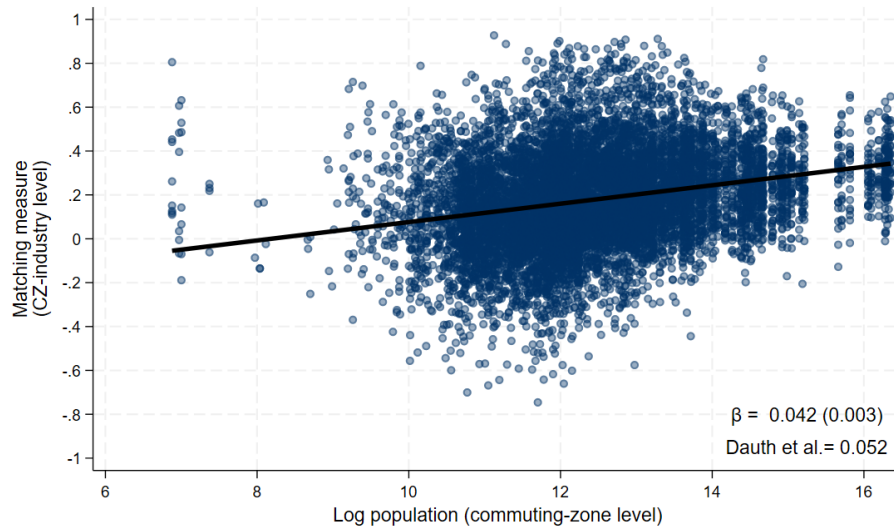
Dependent variable: Matching metric - Correlation of worker and workplace FE		
	(1) CZ	(2) CZ-Industry
A: Baseline Model		
Log Population	0.0456*** (0.006)	0.0460*** (0.003)
R ²	0.117	0.070
B: Correlation of worker and residual workplace FE		
Log Population	0.0415*** (0.005)	0.0460*** (0.003)
R ²	0.113	0.070
C: Log population instrumented with population in 1921-1950		
Log Population	0.0202 (0.027)	0.0567*** (0.017)
R ²	0.088	0.066
First-stage F	4.181	3.022
N for Panels A-C	877	9,160
Mean of dep. variable	0.257	0.188
D: Uncorrected for limited mobility bias		
Log Population	0.0589*** (0.006)	0.0614*** (0.004)
R ²	0.167	0.094
N	877	9,174
Mean of dep. variable	0.092	0.009
E: Dropping the 10% largest areas		
Log Population	0.0454*** (0.006)	0.0438*** (0.004)
R ²	0.112	0.053
N	874	7,888
Mean of dep. variable	0.256	0.199

Source: Author's calculations using IMSS and INEGI data. "CZ" stands for commuting zone. Regressions pool data from 2004-2008, 2009-2013, and 2014-2018 with interval dummies. The estimations exclude CZs with fewer than 50,000 inhabitants. Column (2) restricts to cells with over five firms and 50 workers. Panels: A - baseline ([Bonhomme et al. \(2019\)](#)'s limited-mobility bias correction with five workplace clusters); B - industry-demeaned workplace fixed effects; C - we instrument population with historical population at the Metro and CZ levels, relying on historical population estimates from [Alix-Garcia and Sellars](#). Log population instrumented with log population in 1921, 1930, 1940, and 1950; D - no limited-mobility bias correction; and E - excluding extreme populations. Clustered standard errors at the commuting-zone level in parentheses. *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

H Robustness Checks Excluding Government employees

Figure H.3: City Size and Assortative Matching in Mexico's Formal Labor Markets Excluding Government Employees

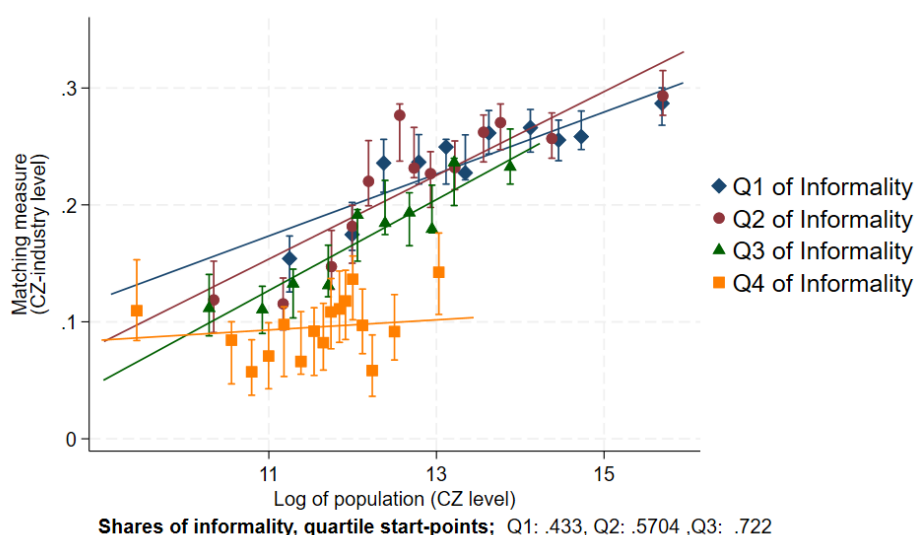
Figure H.4: Commuting Zone - Industry



Source: Author's calculations using IMSS and INEGI data. The figure displays a scatterplot illustrating the relationship between log population and the matching measure estimated from AKM models at the commuting zone industry levels. We exclude government employees and restrict the sample to cells with more than 5 firms and more than 50 workers. We classify industries using 2-digit NAICS codes. The bottom-right value displays the slope of the linear regression for the displayed relationship. The regression includes dummies for each time interval. Clustered standard errors at the commuting-zone level in parentheses.

Figure H.5: City Size and Assortative Matching in Mexico's Formal Labor Markets. Commuting Zone-Industry by Informality Quartiles Excluding Government Employees

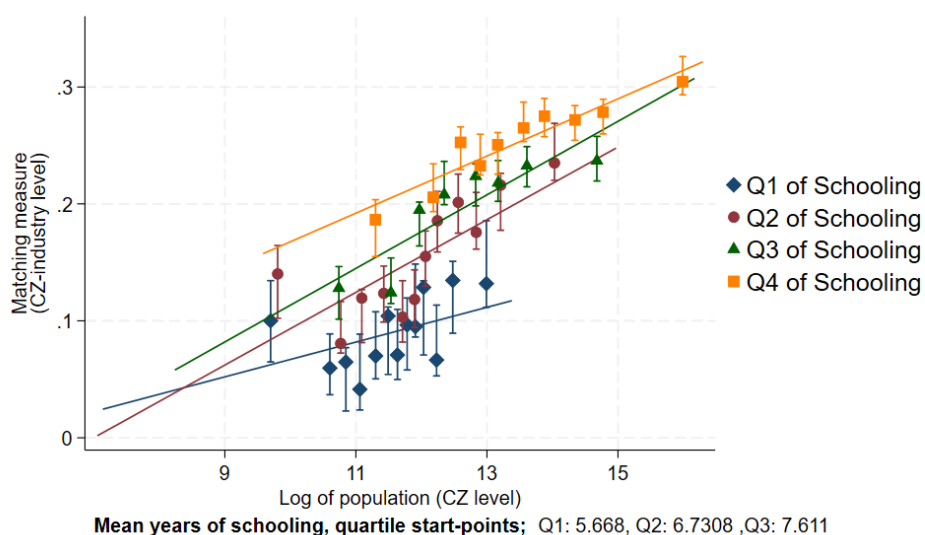
Figure H.6: Commuting Zone - Industry



Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching metric at the commuting zone-industry level for each quartile of the informality rate. We exclude government employees. Informality rates are assigned from the nearest available census: the average of 2000 and 2010 for 2004–2008, 2010 for 2009–2013, and 2015 for 2014–2018. Quartiles are calculated by pooling all years. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

Figure H.7: City Size and Assortative Matching in Mexico's Formal Labor Markets. Commuting Zone-Industry by Years of Schooling Quartiles Excluding Government Employees

Figure H.8: Commuting Zone - Industry



Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and the estimated matching measure at the commuting zone-industry level for each quartile of the mean years of schooling. We exclude government employees. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

I Details about Historical Population Estimates and Additional Instrumental-Variable Results

For Panel C in Table 2, we instrument the population with the historical population at the commuting zone (CZ) level. We use [Alix-Garcia and Sellars' \(2020\)](#) historical population estimates to calculate Mexico's historical population levels. [Alix-Garcia and Sellars \(2020\)](#) divide Mexico's territory into 15x15 km grid cells and estimate the population in each cell from 1900 to 1950. We intersect these grid cells with contemporary municipal boundaries and calculate the historical population of a municipality m as the sum of the populations of each cell that intersects m , weighted by the proportion of m 's land area that intersects each cell. Finally, we aggregate these estimates to the commuting-zone level.

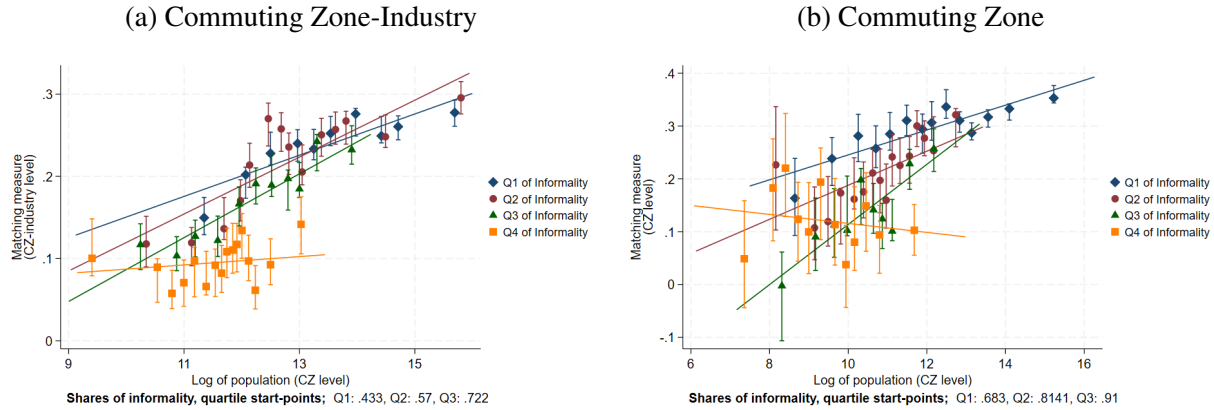
Table I.1: City Size and Assortative Matching in Mexico's Formal Labor Markets Instrumenting Current Population with Historical Population

Dependent variable: Matching metric - Correlation of worker and workplace FE		
	(1)	(2)
	CZ	CZ-Industry
A: Baseline Model		
Log Population	0.0408*** (0.004)	0.0410*** (0.003)
R ²	0.083	0.070
B: Log population instrumented with population in 1921-1950		
Log Population	0.0309*** (0.008)	0.0432*** (0.010)
R ²	0.078	0.070
First-stage F	59.637	4.783
Hansen J statistic	4.521	1.900
p-value Hansen J stat	0.210	0.593
C: Log population instrumented with population in 1930-1950		
Log Population	0.0314*** (0.008)	0.0431*** (0.010)
R ²	0.079	0.070
First-stage F	79.463	6.321
Hansen J statistic	1.074	1.887
p-value Hansen J stat	0.584	0.389
D: Log population instrumented with population in 1940-1950		
Log Population	0.0313*** (0.008)	0.0406*** (0.011)
R ²	0.079	0.070
First-stage F	119.413	4.543
Hansen J statistic	1.059	1.644
p-value Hansen J stat	0.303	0.200
N	1,913	10,097
Mean of dep. variable	0.196	0.180

Source: Author's calculations using data from IMSS, INEGI, and [Alix-Garcia and Sellars \(2020\)](#). The table presents the results of instrumental-variable regressions of the matching measure, using AKM model estimates and log population at the commuting zone and commuting zone-industry levels. We use historical population estimates from [Alix-Garcia and Sellars](#) to calculate Mexico's historical population at the commuting-zone level. See appendix section I for details on the construction of historical population estimates. All the regressions pool data across the three intervals: 2004-2008, 2009-2013, and 2014-2018, and include interval dummies. CZ stands for commuting zone. For column 4, we restrict to cells with more than five firms and more than 50 workers. Panel A shows baseline estimates. First-stage F is the first-stage F-statistic. Clustered standard errors at the commuting-zone level in parentheses. *: p<0.1, **: p<0.05, ***: p<0.01.

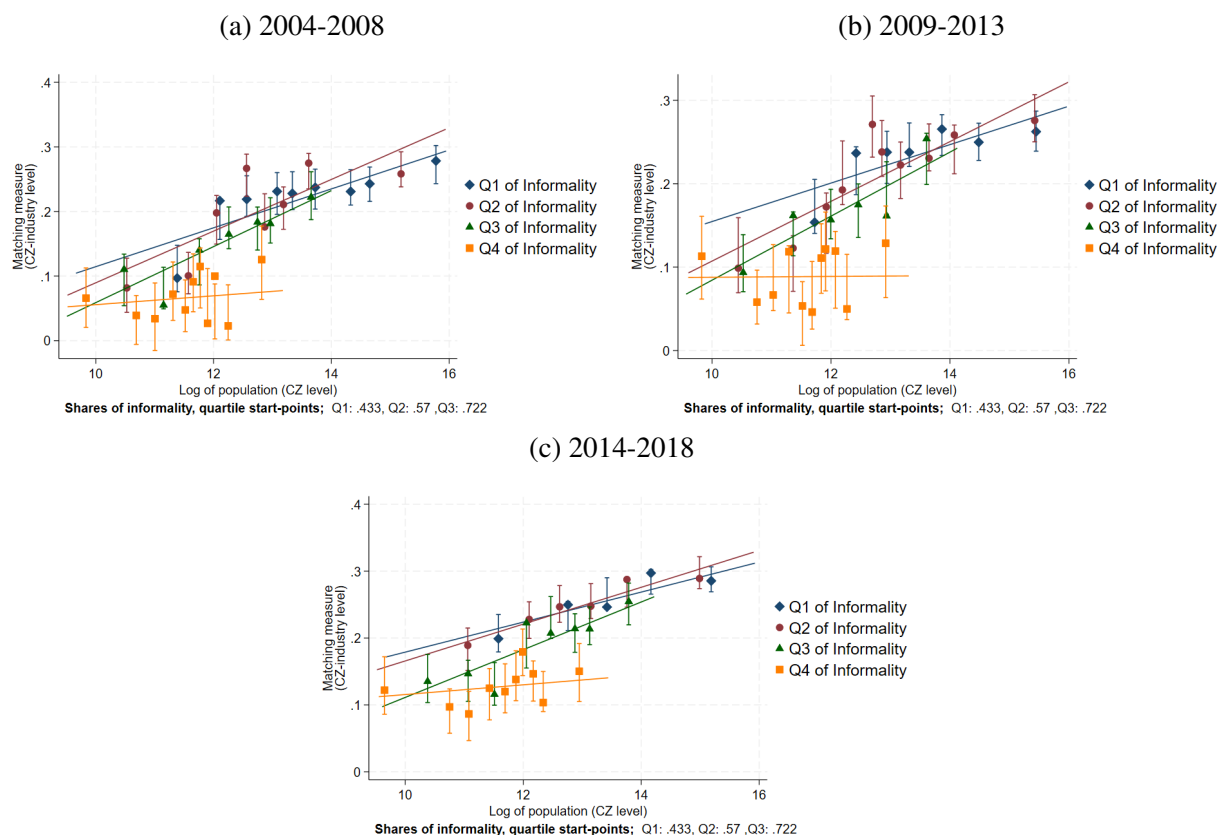
J Additional Results on Mechanisms Affecting Matching Externalities

Figure J.1: City Size and Assortative Matching in Mexico's Formal Labor Markets by Informality Quartiles. Commuting-Zone vs. Commuting-Zone-Sector Level.



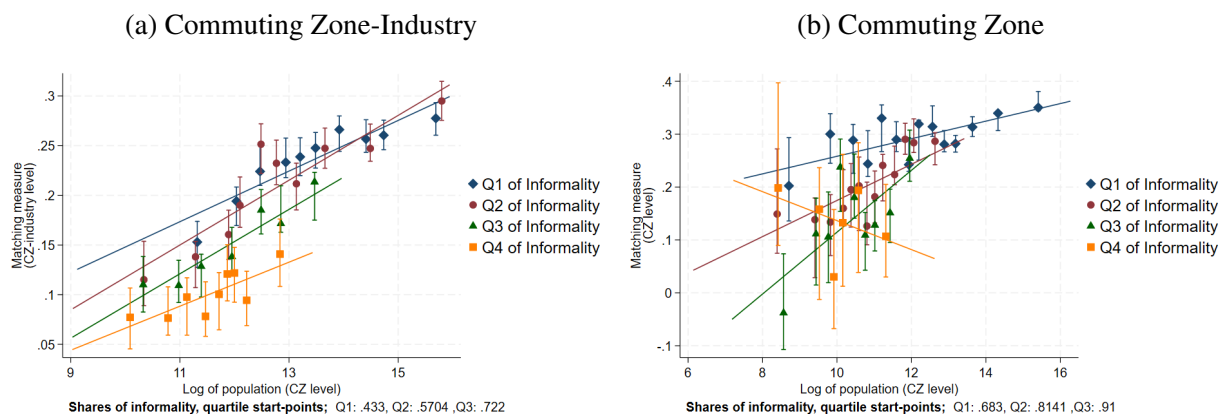
Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching metric at the commuting zone-industry and commuting zone levels for each quartile of the informality rate. For 2004–2008, we use the mean informality rates from the 2000 and 2010 censuses. For 2009–2013, we rely on the 2010 informality rates, and for 2014–2018, we use those based on the 2015 data. Quartiles of the informality rate are calculated over the pooled sample. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

Figure J.2: City Size and Assortative Matching in Mexico's Formal Labor Markets. Commuting Zone-Industry by Informality Quartiles. Estimates by Interval



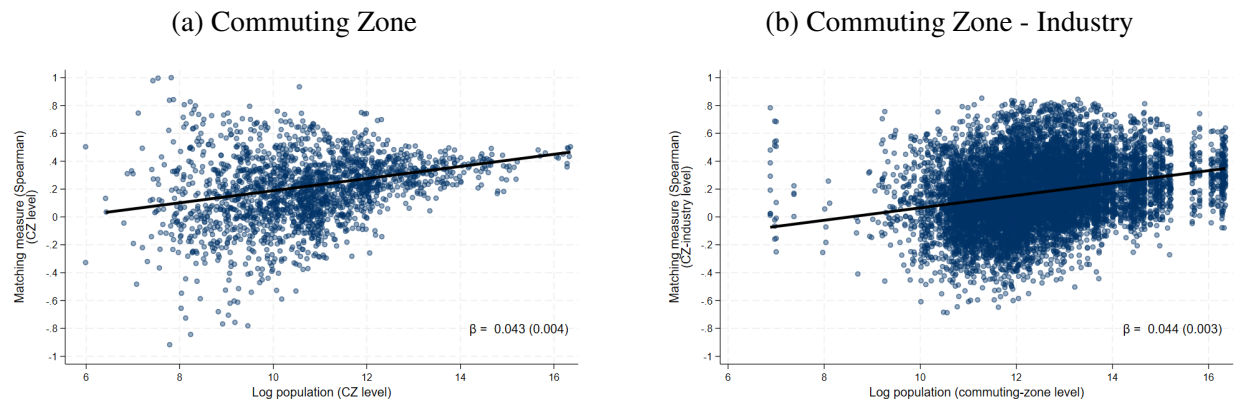
Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching measure at the commuting zone-industry level for each quartile of the informality rate. For 2004–2008, we use the mean informality rates from the 2000 and 2010 censuses. For 2009–2013, we rely on the 2010 informality rates, and for 2014–2018, we use those based on the 2015 data. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plot. The panels correspond to different time intervals.

Figure J.3: City Size and Assortative Matching in Mexico's Formal Labor Markets by Informality Quartiles Excluding the Southern Region.



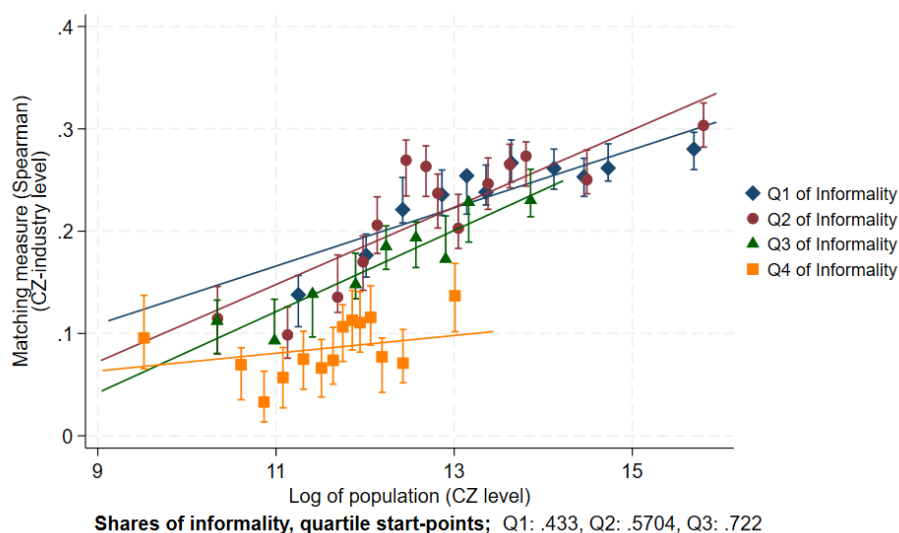
Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching metric at the commuting zone-industry and commuting zone levels for each quartile of the informality rate. For 2004–2008, we use the mean informality rates from the 2000 and 2010 censuses. For 2009–2013, we rely on the 2010 informality rates, and for 2014–2018, we use those based on the 2015 data. Quartiles of the informality rate are calculated over the pooled sample. The sample excludes commuting zones in Mexico's southern region, which comprises the following states: Campeche, Chiapas, Guerrero, Oaxaca, Quintana Roo, Tabasco, Veracruz, and Yucatán. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

Figure J.4: City Size and Assortative Matching in Mexico's Formal Labor Markets: Matching Measured using Spearman Correlation



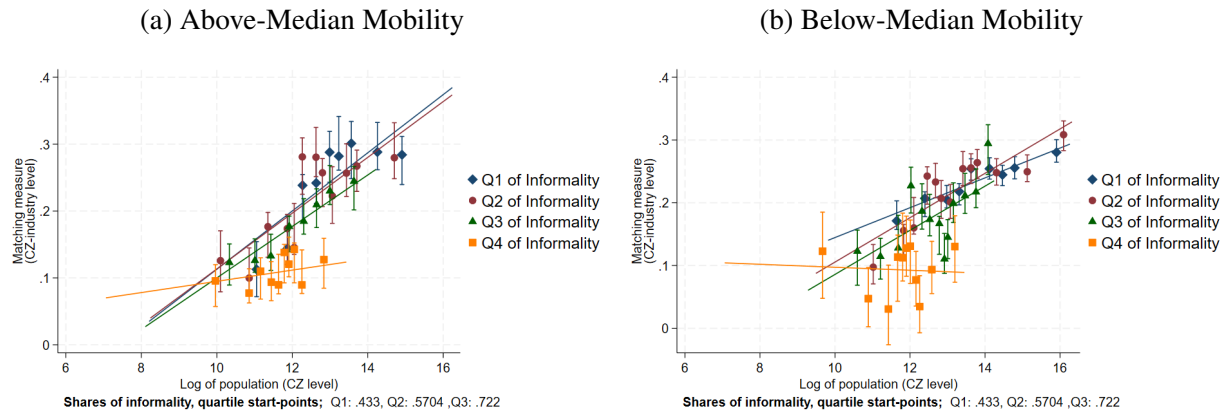
Source: Author's calculations using IMSS and INEGI data. We recalculate the matching measure using Spearman's rank correlation. Each Panel displays a scatterplot illustrating the relationship between log population and the correlation between estimated worker and workplace effects from AKM models at the commuting zone and commuting zone–industry levels. For comparison, Panel (a) displays the relationship estimated for Germany by [Dauth et al. \(2022\)](#). For Panel (b), we restrict to cells with more than five firms and more than 50 workers. We classify industries using 2-digit NAICS codes. The bottom-right values show the slope of the linear regression for the displayed relationship. The regressions include dummies for each time interval. Clustered standard errors at the commuting-zone level in parentheses.

Figure J.5: City Size and Matching in Mexico's Formal Labor Markets. Commuting Zone-Industry by Informality Quartiles. Matching Measured using Spearman Correlation.



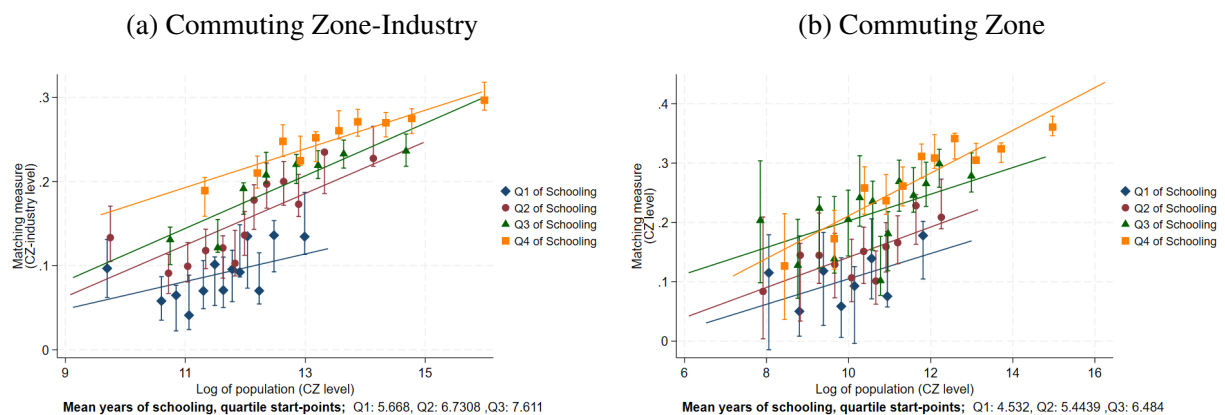
Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of log population and our matching metric at the commuting zone–industry level for each quartile of the informality rate. We recalculate the matching measure using Spearman's rank correlation. Informality rates are assigned from the nearest available census: the average of 2000 and 2010 for 2004–2008, 2010 for 2009–2013, and 2015 for 2014–2018. Quartiles are calculated by pooling all years. The vertical bars show confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

Figure J.6: City Size and Assortative Matching in Mexico's Formal Labor Markets by Informality Quartiles Separating by Worker Mobility



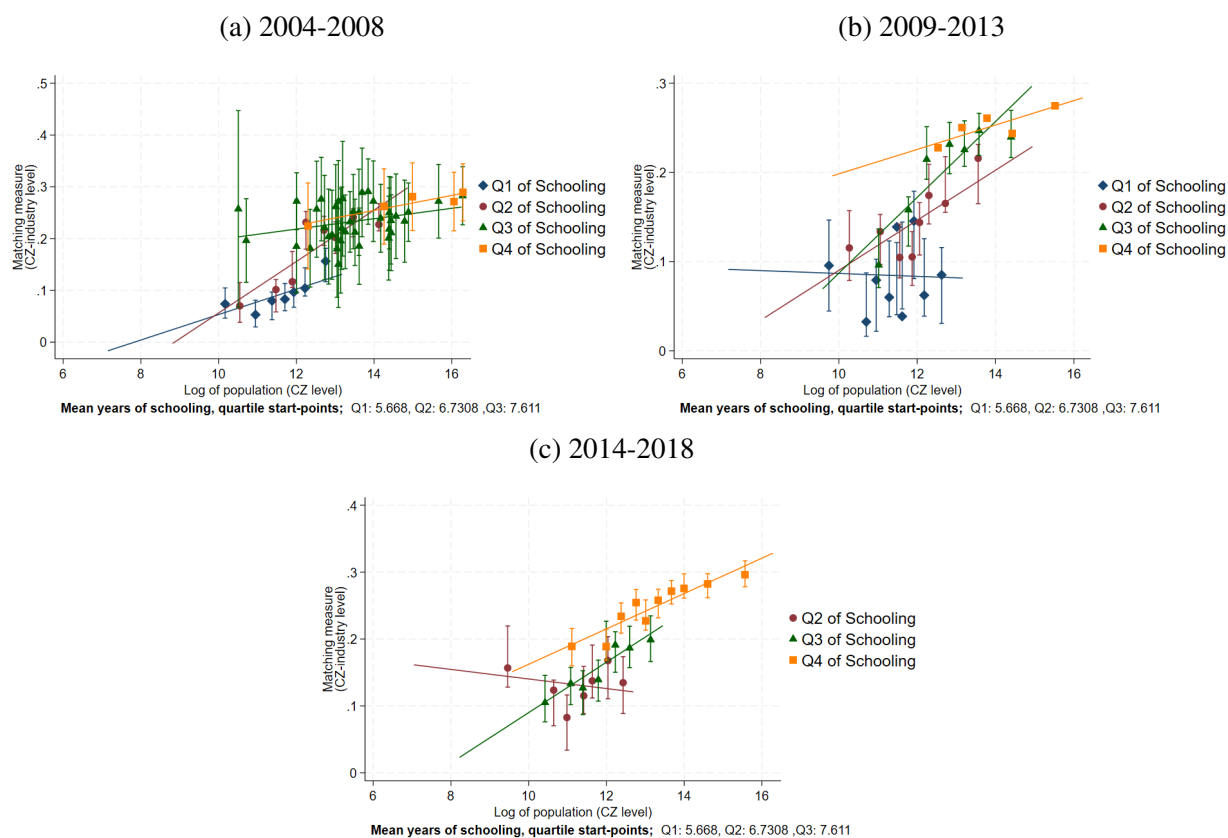
Source: Author's calculations using IMSS and INEGI data. The figures show a binned scatter plot of the log of the population and our matching metric for each quartile of the informality rate. We separate commuting-zone-industry cells based on the eigenvalues of the Laplacian mobility matrix, following [Jochmans and Weidner \(2019\)](#). Panel (a) restricts to commuting-zone-industry cells with above-median mobility. Panel (b) restricts to commuting-zone-industry cells with below-median mobility. For 2004–2008, we use the mean informality rates from the 2000 and 2010 censuses. For 2009–2013, we rely on the 2010 informality rates, and for 2014–2018, we use those based on the 2015 data. Quartiles of the informality rate are calculated over the pooled sample. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands ([Cattaneo et al., 2025, 2024](#)) with default settings to generate the scatter plots.

Figure J.7: City Size and Assortative Matching in Mexico's Formal Labor Markets by Years-of-Schooling Quartiles.



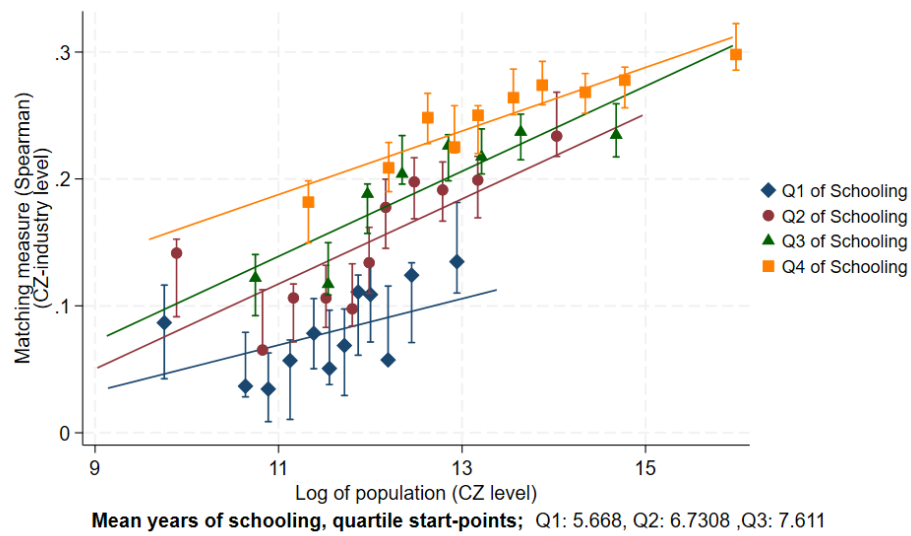
Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log of population and our matching measure at the commuting zone-industry and commuting zone levels, for each quartile of mean years of schooling. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands ([Cattaneo et al., 2025, 2024](#)) with default settings to generate the scatter plots.

Figure J.8: City Size and Assortative Matching in Mexico's Formal Labor Markets. Commuting Zone-Industry by Years-of-Schooling Quartiles. Estimates by Interval



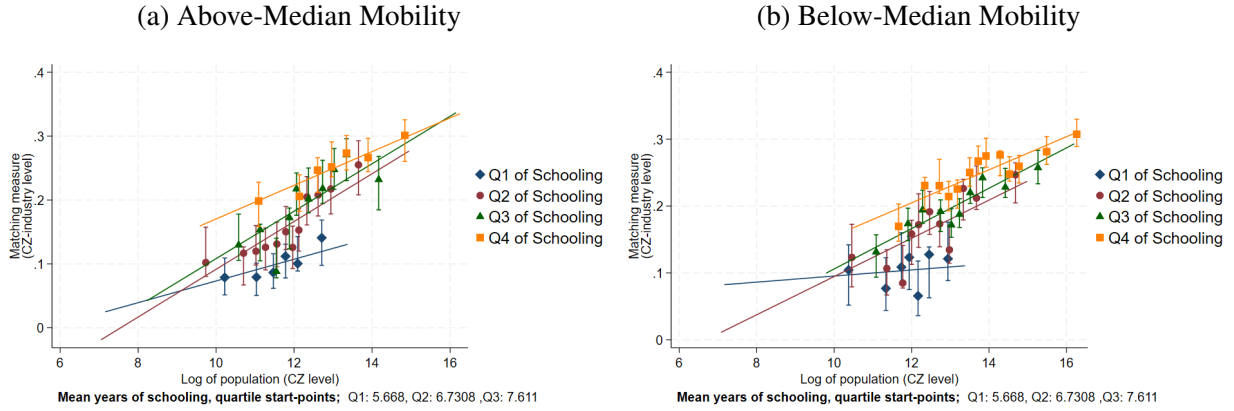
Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching metric at the commuting zone-industry level for each quartile of the mean years of schooling. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots. The panels correspond to different time intervals. Panel (c) excludes the first schooling quartile due to an insufficient effective sample size for estimating the binscatter.

Figure J.9: City Size and Assortative Matching in Mexico's Formal Labor Markets. Commuting Zone-Industry by Years-of-Schooling Quartiles. Matching Measured using Spearman Correlation.



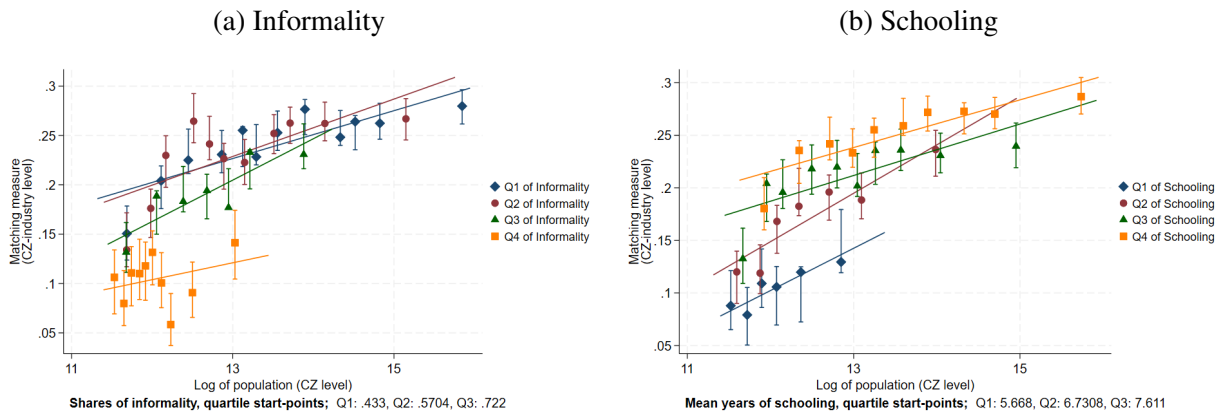
Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of log population and the estimated matching measure at the commuting zone–industry level for each quartile of the mean years of schooling. We recalculate the matching measure using Spearman's rank correlation. The vertical bars represent confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

Figure J.10: City Size and Assortative Matching in Mexico's Formal Labor Markets by Schooling Quartiles Separating by Worker Mobility



Source: Author's calculations using IMSS and INEGI data. The figures display a binned scatter plot of the log population and our matching metric for each quartile of years of schooling. We separate commuting-zone-industry cells based on the eigenvalues of the Laplacian mobility matrix, following [Jochmans and Weidner \(2019\)](#). Panel (a) restricts to commuting-zone-industry cells with above-median mobility. Panel (b) restricts to commuting-zone-industry cells with below-median mobility. Quartiles of schooling are calculated over the pooled sample. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands ([Cattaneo et al., 2025, 2024](#)) with default settings to generate the scatter plots.

Figure J.11: City Size and Assortative Matching in Mexico's Formal Labor Markets. Commuting Zone-Industry by Informality and Mean Years of Schooling Quartiles. Estimates excluding the 10% Largest and Smallest Areas.



Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching measure at the commuting zone-industry level for each quartile of the informality rate (Panel a) and mean years of schooling (Panel b), excluding the 10% largest and smallest areas. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands ([Cattaneo et al., 2025, 2024](#)) with default settings to generate the scatter plots.

Table J.1: City Size and Assortative Matching in Mexico's Formal Labor Markets. Controlling for Informality at the Commuting-Zone-Industry Level and Years of Schooling at the Commuting-Zone Level

Dependent variable: correlation of worker and plant FE				
			Joint estimation	
	(1)	(2)	(3)	(4)
	Informality rate CZ-Industry	Mean years of schooling CZ-Industry	Informality rate CZ-Industry	Mean years of schooling CZ-Industry
Log Pop*first quartile	0.0247*** (0.004)	0.0256*** (0.008)	0.0227*** (0.008)	
Log Pop*second quartile	0.0357*** (0.004)	0.0375*** (0.005)	0.0319*** (0.007)	0.0080 (0.008)
Log Pop*third quartile	0.0225*** (0.004)	0.0309*** (0.003)	0.0170** (0.007)	0.0019 (0.007)
Log Pop*fourth quartile	0.0181** (0.007)	0.0210*** (0.003)	0.0124 (0.010)	-0.0051 (0.008)
Dummy first quartile	-0.0886 (0.056)	-0.1954* (0.099)	-0.0848 (0.097)	
Dummy second quartile	-0.2506*** (0.051)	-0.2847*** (0.061)	-0.2219** (0.084)	-0.0718 (0.092)
Dummy third quartile	-0.1369** (0.050)	-0.1858*** (0.047)	-0.0817 (0.090)	0.0079 (0.090)
Dummy fourth quartile	-0.1579* (0.086)	-0.0228 (0.044)	-0.0967 (0.113)	0.1061 (0.093)
Mean of dep. variable	0.180	0.180	0.180	0.180
R ²	0.476	0.451	0.477	0.477
Obs.	10,091	10,091	10,091	10,091

Source: Author's calculations using IMSS and INEGI data. The table displays the results of regressions of our matching metric and log population, interacted with informality rates at the commuting-zone-industry level and mean years of schooling quartiles at the commuting-zone level. CZ stands for commuting zone. We restrict to cells with more than five firms and more than 50 workers. Clustered standard errors at the commuting-zone level in parentheses.

*: p<0.1, **: p<0.05, ***: p<0.01.

Table J.2: Descriptive Statistics of Workplace and Worker Fixed Effects at the Commuting-Zone-Industry Level by Informality Rate and Mean Years of Schooling Quartiles.

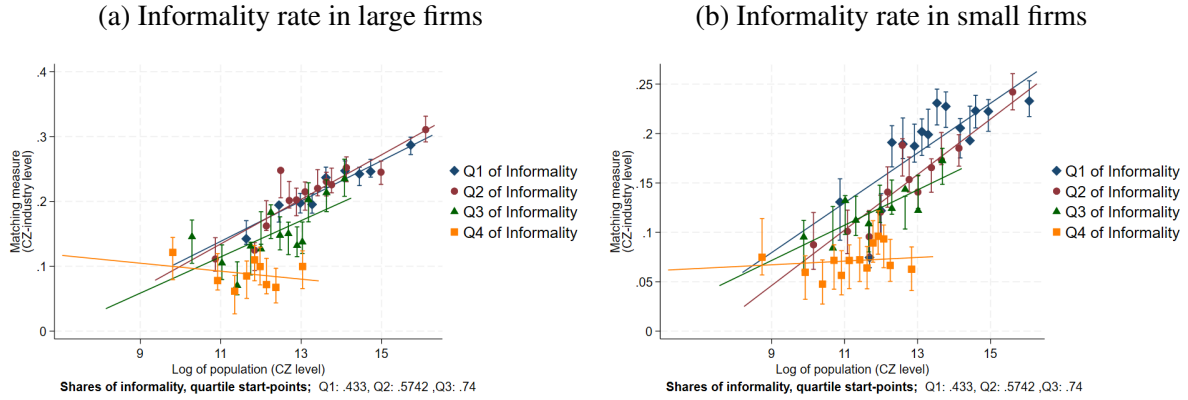
Informality rate	Workplace effects			Worker effects		
	Mean	Mean Variance	IQR	Mean	Mean Variance	IQR
First quartile	-0.284	0.130	0.858	5.739	0.210	0.843
Second quartile	-0.305	0.129	0.874	5.665	0.188	0.950
Third quartile	-0.415	0.119	0.814	5.651	0.161	0.891
Fourth quartile	-0.437	0.104	0.835	5.599	0.147	0.965

Mean years of schooling	Workplace effects			Worker effects		
	Mean	Mean Variance	IQR	Mean	Mean Variance	IQR
First quartile	-0.147	0.102	0.904	5.317	0.152	1.039
Second quartile	-0.379	0.113	1.022	5.624	0.166	0.988
Third quartile	-0.397	0.123	0.868	5.714	0.182	0.623
Fourth quartile	-0.454	0.137	0.458	5.889	0.198	0.399

Source: Author's calculations using IMSS and INEGI data. The table displays descriptive statistics for workplace and worker fixed effects at the commuting zone-industry level, by informality rate and mean years of schooling quartiles. Informality rates are assigned from the nearest available census: the average of 2000 and 2010 for 2004–2008, 2010 for 2009–2013, and 2015 for 2014–2018. Quartiles are calculated by pooling all years. IQR stands for interquartile range.

K Informality and Schooling Across Firm Size

Figure K.1: City Size and Assortative Matching in Mexico's Formal Labor Markets Separating by Informality Rate in Large and Small Firms

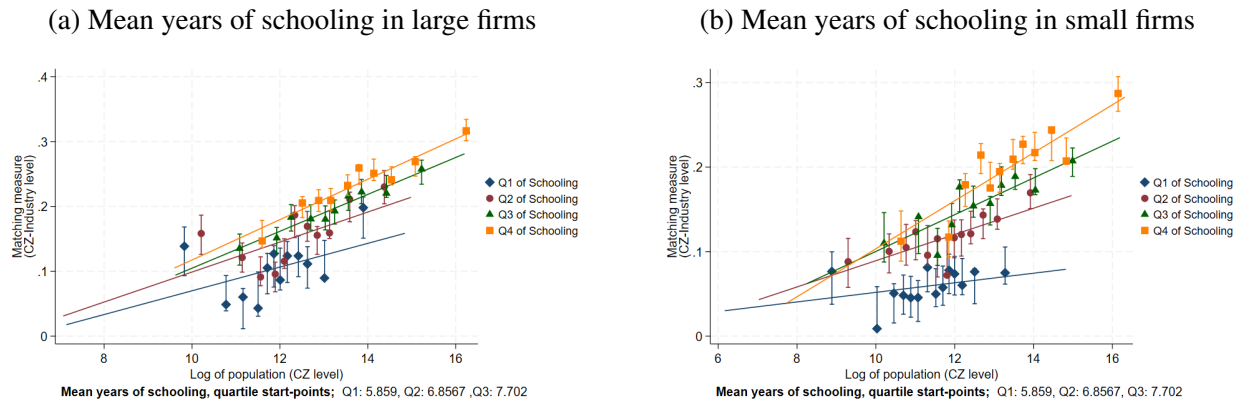


Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching metric at the commuting zone-industry and commuting zone levels for each quartile of the informality rate. Informality rates are assigned from the nearest available census: the average of 2000 and 2010 for 2004–2008, 2010 for 2009–2013, and 2015 for 2014–2018. Quartiles are calculated by pooling all years. Panel (a) shows the relationship separating large firms (16 or more workers) Panel (b) and small firms (fewer than 16 workers). The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

Given the characteristics of the Mexican economy, Figure K.1 demonstrates that city size enhances worker–firm matching more effectively in formal labor markets than in informal ones, with this disparity particularly pronounced among small firms. Across both specifications, the estimated slope of the matching measure with respect to log population varies by informality quartile. The most formal markets (Q1) exhibit a strongly positive relationship, indicating that larger labor markets achieve tighter worker–firm sorting. Conversely, the most informal markets (Q4) display a much flatter slope, suggesting that city size contributes less to improving worker–firm matching in high-informality environments. When the analysis is restricted to small firms, the overall matching measure declines, yet stratification across informality quartiles persists. Markets in Q4 continue to show the lowest matching levels and the weakest relationship with city size. This pattern confirms that poor matching outcomes in high-informality areas cannot be attributed solely to differences in firm size composition.

We also extend this analysis to the schooling level, separating by firm size in Figure K.2. We

Figure K.2: City Size and Assortative Matching in Mexico's Formal Labor Markets Separating by Mean Years of Schooling in Large and Small Firms



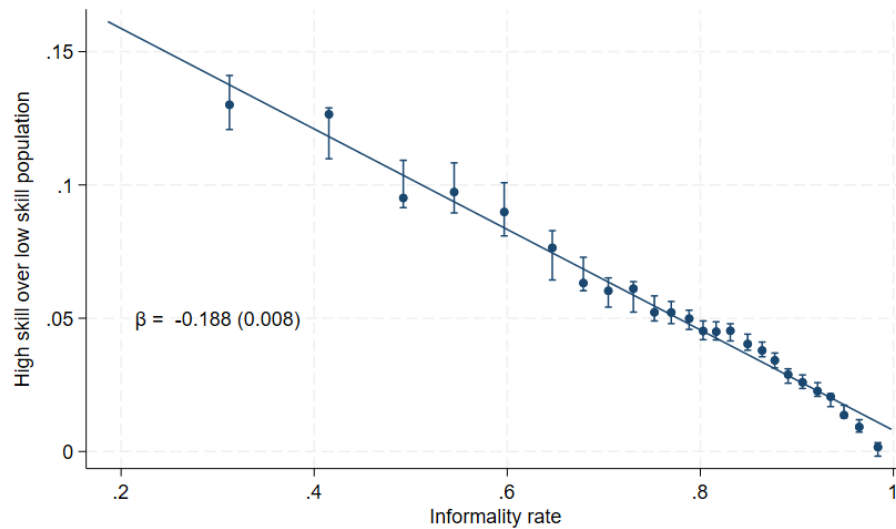
Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and our matching metric at the commuting zone-industry level for each quartile of the mean years of schooling. Panel (a) shows the relationship separating large firms (16 or more workers) Panel (b) and small firms (fewer than 16 workers). The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

exhibit that markets in the top schooling quartile (Q4) exhibit the steepest positive relationship, while those in the bottom quartile (Q1) are nearly flat at low population levels. This supports the view that more educated labor markets achieve greater matching efficiency at the urban scale, consistent with stronger complementarities between skilled workers and firms in larger cities.

The pattern is consistent between large and small firms. In both panels, higher schooling quartiles correspond to higher levels and steeper slopes of the matching measure. The persistence of these results within firm-size groups shows that compositional differences in firm size do not drive the relationship between education and matching efficiency.

L Informality and Skilled Population

Figure L.1: Informality and the Local Supply of Skilled Workers



Source: Author’s calculations using Local Labor Markets in Mexico data from [Aldeco et al. \(2024\)](#). The figure displays a binned scatter plot of the local informality rate (x-axis) and the ratio of high-skill to low-skill population (y-axis) at the CZ level. High-skill refers to individuals with 16 or more years of education, and low-skill to those with fewer than 16 years of education. Vertical bars represent 95% confidence intervals for the conditional mean of the skill ratio at each level of informality. Informality rates are assigned from the nearest available census: the average of 2000 and 2010 for 2004–2008, 2010 for 2009–2013, and 2015 for 2014–2018. Bins are calculated by pooling all years. The slope coefficient from the binned regression is $\beta = -0.188 (0.008, p < 0.001)$. We used the `binsreg` and `binstest` commands ([Cattaneo et al., 2025, 2024](#)) with default settings to generate the scatter plot.

Figure L.1 presents a binscatter of the relationship between the local informality rate and the skill ratio—defined as the share of individuals with at least 16 years of schooling relative to those with lower education levels—constructed from Banco de México’s Local Labor Market data [Aldeco et al. \(2024\)](#). The estimated slope of $-0.188 (0.008)$ indicates a significant negative association: as informality rises, the relative presence of high-skill workers declines sharply.

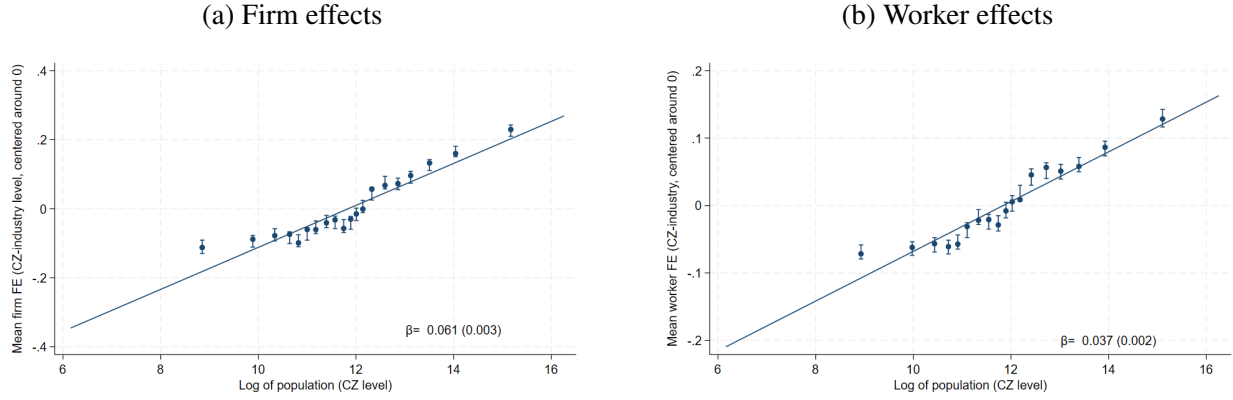
This pooled relationship provides direct “first-stage” evidence supporting the following mechanism: informality undermines the returns to skill and discourages the accumulation or retention of educated labor within local markets. In other words, informal economies sustain a thinner high-skill segment, which limits firms’ ability to find suitably skilled workers and thereby reduces the potential for efficient worker–firm matching.

The evidence in Figure L.1 reinforces the interpretation that the adverse effect of informality

on matching operates in part through its impact on local skill composition: informality depresses human capital accumulation, thereby constraining matching.

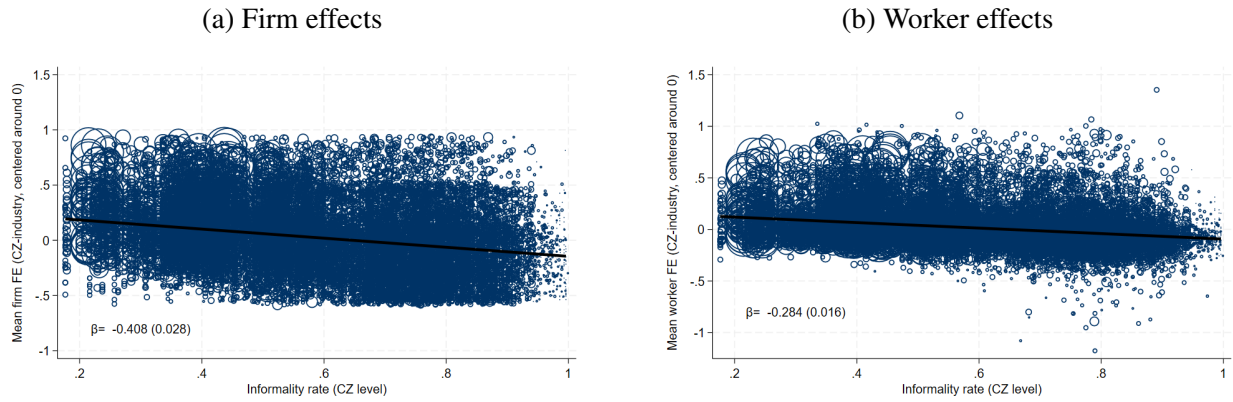
M Co-Location

Figure M.1: Correlation between Worker and Firm Fixed Effects and Log Population CZ-Industry.



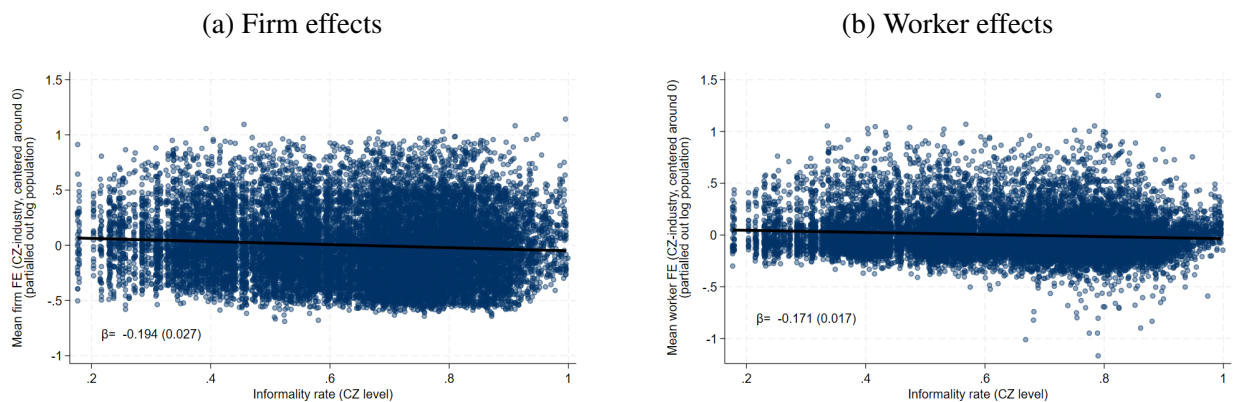
Source: Author's calculations using IMSS and INEGI data. Following [Dauth et al. \(2022\)](#), the figure displays a binned scatter plot of worker and firm fixed effects against the log population. The vertical bars are confidence intervals for the conditional mean of the fixed effects at each population level. We used the `binsreg` and `binstest` commands ([Cattaneo et al., 2025, 2024](#)) with default settings to generate the scatter plots.

Figure M.2: Correlation between Worker and Firm Fixed Effects and Informality Rate at the CZ-Industry Level.



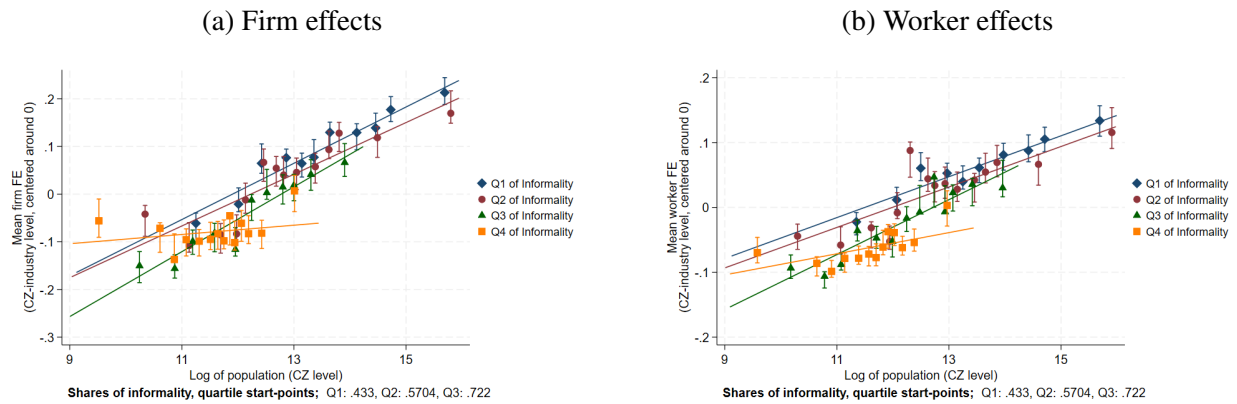
Source: Author's calculations using IMSS and INEGI data. Following [Dauth et al. \(2022\)](#), the figure displays a scatter plot of worker and firm fixed effects against the informality rate, weighted by CZ population.

Figure M.3: Correlation between Worker and Firm Fixed Effects and Informality Rate at the CZ-Industry Level.



Source: Author's calculations using IMSS and INEGI data. Following [Dauth et al. \(2022\)](#), the figure displays a scatter plot of worker and firm fixed effects (partialled out of log population) against the informality rate.

Figure M.4: Informality Rate and Mean Worker and Workplace Fixed Effects.



Source: Author's calculations using IMSS and INEGI data. The figure displays a binned scatter plot of the log population and the mean worker and workplace FE at the commuting zone-industry for each quartile of the informality rate. Informality rates are assigned from the nearest available census: the average of 2000 and 2010 for 2004–2008, 2010 for 2009–2013, and 2015 for 2014–2018. Quartiles are calculated by pooling all years. The vertical bars are confidence intervals for the conditional mean of the correlation at each level of (log) population. We used the `binsreg` and `binstest` commands (Cattaneo et al., 2025, 2024) with default settings to generate the scatter plots.

Appendix References

- Abowd, J. M., Kramarz, F., and Margolis, D. N. (1999). High Wage Workers and High Wage Firms. *Econometrica*, 67(2):251–333
- Aldeco, L., Calderón, M., Chiquiar, D., Hanson, G., Pérez Pérez, J., and Velázquez, C. (2024). Local Labor Markets in Mexico: Definition, Databases, and Descriptive Analysis
- Alix-Garcia, J. and Sellars, E. A. (2020). Locational Fundamentals, Trade, and the Changing Urban Landscape of Mexico. *Journal of Urban Economics*, 116:103213
- Bonhomme, S., Lamadon, T., and Manresa, E. (2019). A Distributional Framework for Matched Employer Employee Data. *Econometrica*, 87(3):699–739
- Card, D., Heining, J., and Kline, P. (2013). Workplace Heterogeneity and the Rise of West German Wage Inequality. *The Quarterly Journal of Economics*, 128(3):967–1015
- Cadena, B. (2014). Recent Immigrants as Labor Market Arbitrageurs: Evidence from the Minimum Wage. *Journal of Urban Economics*, 80(C):1–12
- Fowler, C. S. and Jensen, L. (2020). Bridging the Gap Between Geographic Concept and the Data We Have: The Case of Labor Markets in the USA. *Environment and Planning A: Economy and Space*, 52(7):1395–1414
- Kline, P., Saggio, R., and Sølvesten, M. (2020). Leave-out Estimation of Variance Components. *Econometrica*, 88(5):1859–1898
- Pérez Pérez, J. and Nuño-Ledesma, J. G. (2024). Workers, Workplaces, Sorting, and Wage Dispersion in Mexico. *Economía LACEA Journal*, 23(1)